



BACHELOR OF SCIENCE IN ELECTRONIC AND TELECOMMUNICATION ENGINEERING

Dual-band Metamaterial Absorber for C and X Band Applications

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February, 2025

DEDICATION

This dissertation is a dedication to my worthy parents, educators, and the Martyrs of July.

CERTIFICATION OF APPROVAL

The Department of Electronic and Telecommunications Engineering (ETE) at the International Islamic University Chittagong (IIUC) has recognized the thesis study of Md Mujahid Hossain (ID No. T201046) titled "**Dual-band Metamaterial Absorber for C and X Band Applications**" as adequate to meet the requirements for earning a Bachelor's degree in Electronic and Telecommunications Engineering.

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CANDIDATE DECLARATION

The material contained in this thesis has not been previously submitted for credit in pursuit of any other degrees or diplomas, nor does it include any information that could be interpreted as breaching any current regulations in force.

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ACKNOWLEDGEMENTS

I am grateful to Allah (SWT) and the Prophet Mohammed (SAAW), whose guidance and blessings have illuminated my path. Their wisdom and mercy have consistently provided strength, shaping my journey with steadfast faith and purpose. In humble recognition, I dedicate my efforts to their divine influence, seeking blessings and guidance at every step. This dedication is a testament to my unwavering faith and commitment.

First and foremost, I am thankful to Allah (SWT) for enabling us to complete my research with the capability and determination to do so. His divine guidance and blessings have been instrumental in this journey. Secondly, I want to thank my thesis supervisor, Dr. Saif Hannan, for his exceptional guidance and unwavering motivation throughout the research process. I also appreciate the Convener, Engr. Mohammed Jashim Uddin, for their ongoing support and inspiration during my journey. Their technical insights and vision have contributed significantly to the advancement of this thesis. I further extend my gratitude to the academic committee, whose feedback and suggestions have played a crucial role in shaping this work.

Moreover, I recognize and value the steadfast support of my parents, who have been with me through every stage of my life. I sincerely thank my family, friends, and well-wishers for their direct or indirect involvement in completing this thesis. Their emotional support and encouragement, which have been a continuous source of strength for which I am profoundly thankful.

ABSTRACT

This research emphasizes the design and implementation of a dual-band metamaterial absorber (MMA) tailored for the C and X bands. The primary objective is to enhance the performance of enhanced mobile broadband (eMBB) within the 5G n79 band, focusing on a resonant frequency of 5 GHz, while also facilitating remote sensing of soil salinity in the 5.3 GHz C-band. The MMA showcases impressive absorption capabilities, achieving high absorption rates of 97.49% at 5.016 GHz, 95.88% at 5.304 GHz, 91.70% at 7.544 GHz, 99.92% at 8.32 GHz, and 97.14% at 10.92 GHz, all within a bandwidth of 86 MHz, specifically between 4.972 GHz and 5.058 GHz, making it ideal for 5G n79 applications. By incorporating this dual-band absorber, we assess its ability to enhance signal quality and minimize interference in eMBB contexts, ultimately boosting the efficiency of the network. Moreover, the MMA presents additional resonant frequencies at 5.304 GHz (62 MHz), 7.544 GHz (47 MHz), 8.32 GHz (143 MHz), and 10.92 GHz (136 MHz), which broadens its potential uses in various communications systems, including IoT connectivity and radar technologies, emphasizing its adaptability in contemporary wireless communication settings.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
RCS	Radar Cross-Section
IEEE	Institute of Electrical and Electronics Engineers
Wi-Fi	Wireless Fidelity
LTE	Long Term Evolution
WLAN	Wireless Local Area Network
LAN	Local Area Network
IoT	Internet of Things
MIMO	Multiple Input Multiple Output
Gbps	Gigabits per Second
MM	Metamaterial
MMA	Metamaterial Absorber
PMA	Perfect Metamaterial Absorber
SRR	Split-Ring Resonator
CSRR	Concentric Split-Ring Resonator
PBG	Photonic Bandgap

ZIM	Zero Index Metamaterial
S-parameter	Scattering Parameter
ADS	Advanced Design System
CST	Computer Simulation Technology
dB	Decibel
E-field	Electric Field
EM	Electromagnetic
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EMR	Effective Medium Ratio
ENG	Epsilon Negative
SNG	Single Negative
DNG	Double Negative
DPS	Double Positive
MNG	Mu-negative
ITO-FET	Indium Tin Oxide-Polyethylene Terephthalate Field-Effect Transistor

LCR	Inductance-Capacitance-Resistance
DRI	Direct Refractive Index
FIU	Filling Factor
FFS-CSRA	Four-Fold Symmetric Circular Split-Ring Absorber
CRECGR	Couple Ring Enclosed Circular Geometric Resonator
BSM	Blind Spot Monitoring
SAR	Specific Absorption Rate
DB-WS	Dual Band Wi-Fi Signal
DB-WSA	Dual Band Wi-Fi Signal Absorber
TM	Transverse Magnetic
TE	Transverse Electric
BW	Backward wave
5G	Fifth Generation
(NIM)	Negative Index Materials
eMBB	Enhanced Mobile Broadband
LH	Left-handed
SAR	Specific Absorption Rate

RF	Radio Frequency
MIMO	Multiple Input Multiple Output
mm	Millimeter
GHz	Giga Hertz
DPS	Double Positive
Wi-Fi	Wireless fidelity
H-Field	Magnetic Field
FR4	Flame Retardant 4

CHAPTER 1

INTRODUCTION

1.1 Advancement in the field of metamaterials

Metamaterials are engineered materials that possess unique electromagnetic properties not found in naturally occurring substances [1]. These materials are structured at a scale smaller than the wavelength of the electromagnetic radiation they interact with, allowing them to manipulate waves in ways that conventional materials cannot. One of the most significant applications of metamaterials is in the development of metamaterial absorbers (MMAs), which are designed to efficiently absorb electromagnetic radiation across specific frequency ranges, including microwave, terahertz, and optical wavelengths [2],[3]. The concept of a metamaterial absorber emerged from the need for materials that can minimize reflection and maximize absorption of incident electromagnetic waves. MMAs have garnered attention for their potential applications in various fields such as stealth technology, solar energy harvesting, sensors, and photodetectors, Shown in Figure 1. They can be particularly effective in applications requiring high absorption rates and low reflection coefficients, making them ideal for enhancing performance in wireless communication systems. A dual-band metamaterial absorber is a specialized type of MMA that can absorb electromagnetic radiation at two distinct frequency bands simultaneously. This capability is achieved through the careful design of the metamaterial structure, which incorporates different resonant elements tailored to respond to specific wavelengths. The ability to absorb energy at multiple frequencies is crucial for applications in satellite communication and radar systems, particularly within the C (4-8 GHz) and X (8-12 GHz) bands. The C and X bands are critical for various modern technologies, including enhanced mobile broadband (eMBB) services in 5G networks and remote sensing applications such as soil salinity detection [3]. The increasing demand for efficient communication systems necessitates advancements in materials that can effectively manage electromagnetic interference and improve signal quality [4].

In this context, the development of a dual-band metamaterial absorber specifically engineered for C and X band applications represents a significant step forward. By leveraging the unique properties of metamaterials, this research aims to create an MMA

that not only meets the stringent requirements for absorption efficiency but also addresses challenges such as size constraints, polarization sensitivity, and bandwidth limitations inherent in existing designs. Overall, the exploration of dual-band MMAs opens new avenues for innovation in wireless communication technologies, paving the way for enhanced performance in diverse applications ranging from telecommunications to environmental monitoring [5].

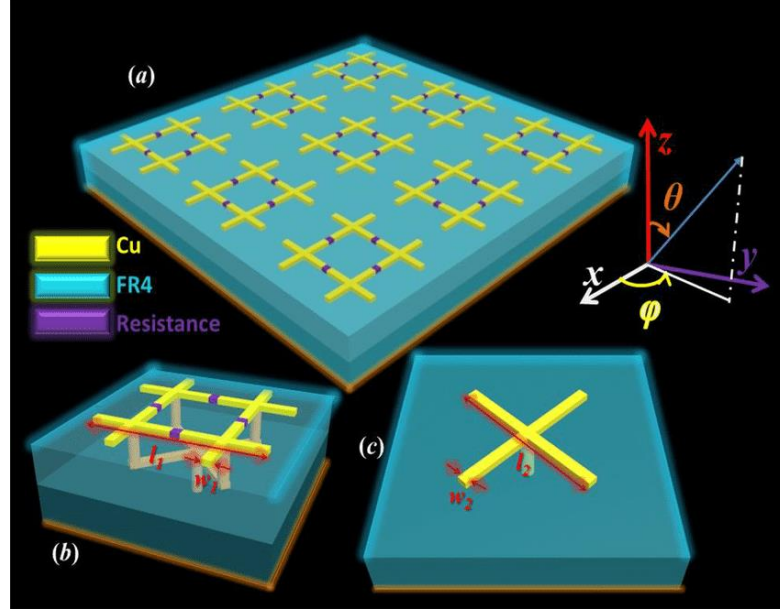


Figure 1. Metamaterial Absorber [6].

1.2 Metamaterials

Metamaterials are artificially structured materials engineered to exhibit properties not typically found in natural materials. They are composed of periodic arrangements of unit cells that interact with electromagnetic waves in unique ways, allowing for the manipulation of wave propagation, absorption, and scattering. The defining characteristic of metamaterials is their ability to achieve a negative index of refraction, enabling phenomena such as superlensing and cloaking. These materials can be designed to operate across a wide range of frequencies, including microwave, terahertz, and optical wavelengths. Metamaterials have garnered significant interest in various applications, including telecommunications, sensing, imaging, and stealth technology.

In particular, metamaterial absorbers (MMAs) have emerged as a crucial area of research due to their capability to absorb incident electromagnetic radiation efficiently across specific frequency bands. By tailoring the geometry and composition of the

metamaterial structure, researchers can achieve high absorption rates while minimizing reflection and transmission losses. This versatility makes metamaterials an essential component in advancing modern technologies, particularly in enhancing the performance of wireless communication systems and other electromagnetic applications.

1.3 Metamaterial absorber (MMA)

A metamaterial absorber (MMA) is a specially designed artificial structure that can absorb electromagnetic waves effectively over specific frequency ranges. Unlike conventional materials, MMAs utilize engineered geometries and periodic patterns to manipulate electromagnetic properties such as permittivity and permeability. These absorbers are primarily characterized by their ability to minimize reflection and transmission, leading to high absorption of incident electromagnetic energy, as shown in Figure 2 [7].

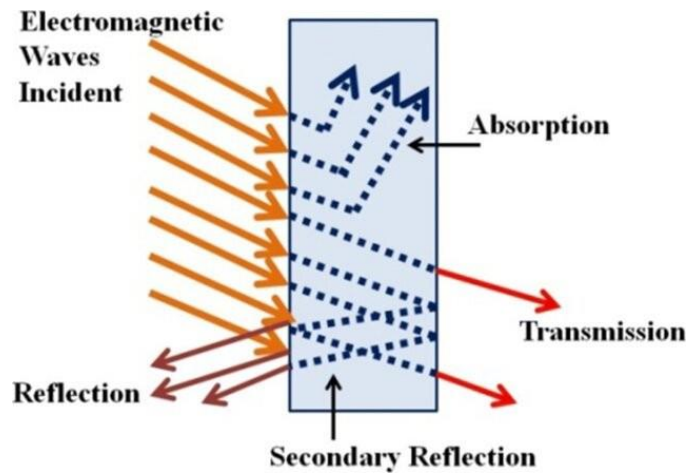


Figure 2. Interaction of electromagnetic waves with the MMA [7].

1.4 Perfect metamaterial absorber (PMA)

A Perfect Metamaterial Absorber (PMA) is an advanced class of metamaterial designed to achieve near-complete absorption of incident electromagnetic waves at specific frequencies. This remarkable performance is achieved through the precise tuning of the metamaterial's permittivity and permeability, resulting in zero reflection and negligible transmission. PMAs utilize subwavelength resonators and impedance matching to free space, ensuring optimal absorption [7],[8]. Their unique properties make them highly effective for applications requiring maximum energy attenuation, such as radar stealth

technology, wireless communication systems, and high-sensitivity sensors. PMAs have also found increasing utility in energy harvesting and imaging technologies due to their capability to focus electromagnetic energy with minimal losses. Their unique properties make them highly effective for applications requiring maximum energy attenuation, such as radar stealth technology, wireless communication systems, and high-sensitivity sensors [9].

1.5 Characteristics of metamaterial absorber

Metamaterials can be characterized by various factors that define their electromagnetic properties. The key factors that encompass these properties include:

1.5.1 Electrostatic field (E-field)

In the field of metamaterials, the electric field plays a crucial role in shaping their electromagnetic properties. The interaction between the electric field and the subwavelength structures or elements within the metamaterial results in unique and precisely tailored electromagnetic characteristics [9].

The idea of the electric field originates from the influence of nearby electric charges and is a core principle in electromagnetism. Represented as 'E,' the electric field is a vector field defined by both its strength and direction. It arises from the presence of electric charges, with positive charges radiating electric fields outward and negative charges projecting fields inward. The intensity and polarity of surrounding charges directly influence the strength of the electric field at a particular location in space. The electric field is fundamental to the study of electricity and magnetism, offering insights into the interactions and behaviors of electric charges in various contexts. Its importance is apparent in many practical applications, such as the creation of electrical grids, functioning of computer hardware, and the generation and distribution of electricity [10]. Within metamaterials, electric fields significantly affect their electromagnetic traits. This impact allows for the deliberate adjustment of the material's response to electric fields across different frequencies, facilitating the attainment of desired attributes like a negative refractive index, improved absorption, or the precise tuning of resonant behavior [10]. Surface currents can arise from several factors, including electromagnetic radiation, induced currents, or deliberate design in devices like antennas and microwave equipment. This deliberate design aspect highlights the

intentionality and control that can be exerted in managing surface currents, which is crucial across various domains, including electromagnetics, electronics, and telecommunications.

1.5.2 Magnetic field (M field)

In the realm of metamaterials, "magnetic field" describes the area surrounding the metamaterial where magnetic influences and interactions take place. Engineers design the configuration of subwavelength structures in these materials to manipulate and enhance their magnetic characteristics, enabling customized responses to magnetic fields [8],[10].

A magnetic field, which is a key concept in electromagnetism, relates to the effect that adjacent magnetic or electric fields have on a magnetic charge or a moving electric charge. Represented by the symbol 'B,' this concept is characterized as a vector field, having both direction and magnitude. Magnetic fields can arise from magnetic charges, known as poles, or from the movement of electric charges. In cases where magnetic charges are absent, these fields are produced by the movement of electric charges, such as electrons flowing through a conductor. The direction of the magnetic field at any given point in space is determined by the flow of current or by the arrangement of magnetic poles [10].

1.5.3 Surface current

Surface current, the movement of electric charge along the exterior of a conductor or material, is a concept that can be deliberately harnessed. It occurs when electric charges, usually electrons, traverse the surface of a material, such as metal or a dielectric, instead of moving through the substance itself [10]. Surface currents can arise from several factors, including electromagnetic radiation, induced currents, or deliberate design in devices like antennas and microwave equipment. This deliberate design aspect highlights the intentionality and control that can be exerted in managing surface currents, which is crucial across various domains, including electromagnetics, electronics, and telecommunications.

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can be exerted in managing surface currents, which is crucial across various domains, including electromagnetics, electronics, and telecommunications [9],[10].

1.5.4 Transmission

Metamaterial absorbers are engineered to effectively capture particular frequencies or bands of electromagnetic energy. Consequently, these absorbers are expected to show very little transmission at the targeted absorption frequency or within the absorption band. In simpler terms, they are anticipated to greatly diminish or entirely obstruct incoming electromagnetic radiation within the specified absorption frequency or range [11]. Nonetheless, based on the absorber's design, it may still permit some transmission of incoming radiation at frequencies that fall outside the absorption range. This indicates that while the metamaterial absorber primarily aims to block specific frequencies, it does not fully eliminate the possibility of transmission. The degree of transmission can be fine-tuned by altering the structural parameters of the absorber [11].

1.5.5 Reflectance

Metamaterial absorbers are crafted to improve absorption by limiting the reflection of electromagnetic energy within specific frequency bands [12]. Their effectiveness relies on several factors, including the properties of the substrate, the width of the absorber, and the size of the individual unit cells. In theory, enhancing absorption involves decreasing reflectivity, achievable by designing absorbers that perform efficiently at targeted frequencies. The correct placement of the absorber is vital to avoid unwanted reflections or scattering, which could negatively affect the system's performance. To evaluate the effectiveness of a metamaterial absorber, it is important to conduct simulations and experimental assessments at various frequencies and incident angles. By modifying parameters such as the size of the unit cells, the thickness of the absorber, and the characteristics of the substrate, one can reduce reflectivity while maintaining high absorption capability [9]-[12].

1.5.6 Absorption

The phrase 'metamaterial absorber absorption' encapsulates the potential for customization in materials science. It signifies the ability of a metamaterial structure to capture electromagnetic waves within a specific frequency range [13]. These metamaterial absorbers are meticulously designed using engineered materials with

unique electromagnetic characteristics not found in conventional materials. Their primary aim is to enhance absorption efficiency within a targeted frequency band. Factors such as the size and composition of the unit cells, the absorber's width, and the substrate's properties are all variables that can be manipulated to influence how well a metamaterial absorber can absorb electromagnetic radiation. By adjusting these parameters, it is possible to create selective absorbers that focus on specific frequencies or improve absorption efficiency over a broader range of frequencies, giving researchers and engineers a powerful tool for their work [14].

Practical methods such as measuring reflectance and transmittance can be employed to experimentally assess the absorption efficiency of a metamaterial absorber, providing crucial insights into the absorber's performance. The absorption coefficient, often expressed numerically, serves as a clear indicator of the proportion of incoming electromagnetic radiation the absorber successfully captures. Metamaterial absorbers find applications in a myriad of fields, from sensing and imaging to energy harvesting. In these areas, the effective absorption of electromagnetic radiation within a specified frequency range is not just a goal, but a necessity, and metamaterial absorbers are at the forefront of making this a reality [14]. Practical methods such as measuring reflectance and transmittance can be employed to experimentally assess the absorption efficiency of a metamaterial absorber, providing crucial insights into the absorber's performance.

1.5.7 Relative permittivity (ϵ_r)

Permittivity is an intrinsic property of materials that measures their resistance to the formation of an electric field. Represented by the Greek letter ϵ , it indicates the amount of charge needed to produce one unit of electric flux within a particular medium. This characteristic is vital for comprehending and describing how materials interact with electric fields. The lowest possible permittivity occurs in a vacuum, which is commonly referred to as the Permittivity of Free Space or the electric constant, symbolized as ϵ_0 , with a numerical value of 8.85×10^{-12} Farad/meter.

Even in insulating materials, there is a noticeable resistance to the formation of electric field lines. On the other hand, relative permittivity refers to the ratio of a dielectric's absolute permittivity to the electric constant, illustrating its permittivity in comparison to that of a vacuum [14]. The speed of electromagnetic waves is influenced by the optical density of the medium, which reflects its ability to absorb and release

electromagnetic energy. A substance with a higher optical density will slow down the speed of light more than a substance with a lower optical density

It is a unitless measure and is expressed as:

$$\text{Permittivity} = \frac{\text{Displacement of Electric}}{\text{Intensity of Electric Field}} \quad (1.1)$$

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} \quad (1.2)$$

where ϵ_0 represents the permittivity of free space, ϵ_r is the dielectric constant, and ϵ denotes the intrinsic permittivity of that substance [15].

1.5.8 Permeability (μ)

Magnetic permeability refers to the characteristic of a material that enables the establishment of magnetic lines of force or a magnetic field. It indicates how well a material can become magnetized when exposed to an external magnetic field [16]. Simply put, magnetic permeability can be explained as "the extent to which a substance permits magnetic field lines to pass through" or "the ability of a material to direct magnetic field lines." This property is commonly denoted by the Greek letter μ [16].

$$\text{Permittivity} = \frac{\text{Magnetic induction of Magnituide (B)}}{\text{Strength of the magnetic field(H)}} \quad (1.3)$$

$$\mu = \frac{B}{H} \quad (1.4)$$

Magnetic permeability describes a material's ability to support the formation of magnetic lines of force or create a magnetic field. It reflects how effectively a material can be magnetized when subjected to an external magnetic field. In simpler terms, magnetic permeability can be understood as "the degree to which a substance allows magnetic field lines to penetrate" or "the capacity of a material to guide magnetic field lines." This characteristic is typically represented by the Greek letter μ [16].

1.5.9 Refractive index

The concept of the refractive index, also known as the index of refraction, is surprisingly simple. It's just the ratio between the speed of light in a vacuum and the speed of light in a specific medium. The properties of the medium determine the velocity of light as it passes through [17]. The speed of electromagnetic waves is

influenced by the optical density of the medium, which reflects its ability to absorb and release electromagnetic energy. A substance with a higher optical density will slow down the speed of light more than a substance with a lower optical density [15],[16].

The refractive index, a dimensionless quantity, is a key indicator of a medium's optical density. It's not just a theoretical concept-it's used in a wide range of practical applications. It represents how much slower light travels in that medium compared to its speed in a vacuum. Represented as 'n,' the refractive index is the ratio of the speed of light in a vacuum to the speed of light in the specific medium.

The equation for the refractive index is as follows [17]:

$$n = \frac{c}{v} \quad (1.5)$$

Here, 'n' represents the refractive index, 'c' denotes the speed of light in a vacuum (3×10^8 m/s), and 'v' stands for the speed of light in a material.

1.6 Satellite communication

Satellite communication involves the transmission of information, data, or signals between two or more locations using communication satellites in Earth's orbit. This technology facilitates long-distance communication and is commonly employed for various purposes, such as television broadcasting, internet connectivity, telephone calls, and data transfer. Satellite communication systems typically consist of ground stations, uplink facilities, satellites in space, and downlink stations [18]. Ground stations send data to satellites, which then relay this information to other locations on Earth or to additional satellites for further distribution. This technology is particularly valuable for connecting remote or geographically isolated areas and is widely utilized in global telecommunications, broadcasting, military and defense applications, navigation systems, and more, as shown in Figure 3.

Satellite communication has revolutionized global connectivity, providing a reliable and efficient means of bridging vast distances and overcoming geographical barriers. Unlike traditional wired communication systems, satellites enable real-time data exchange between distant locations without the need for extensive physical infrastructure on the ground. This makes satellite communication indispensable in areas where terrestrial networks are challenging to deploy, such as mountainous regions,

remote islands, and disaster-stricken zones where existing networks may be disrupted. Moreover, satellite networks play a critical role in supporting maritime and aviation communication, ensuring uninterrupted connectivity for ships and aircraft navigating across oceans and skies. By facilitating high-speed internet access, satellite communication has also contributed to reducing the digital divide in underserved communities, promoting social and economic inclusion. Advancements in satellite technology, such as the development of geostationary, medium-Earth orbit (MEO), and low-Earth orbit (LEO) satellites, have further enhanced the capacity, speed, and reliability of communication networks. These technological strides have broadened the scope of satellite applications, supporting emerging technologies like the Internet of Things (IoT), telemedicine, and smart agriculture, thereby transforming industries and driving global development.

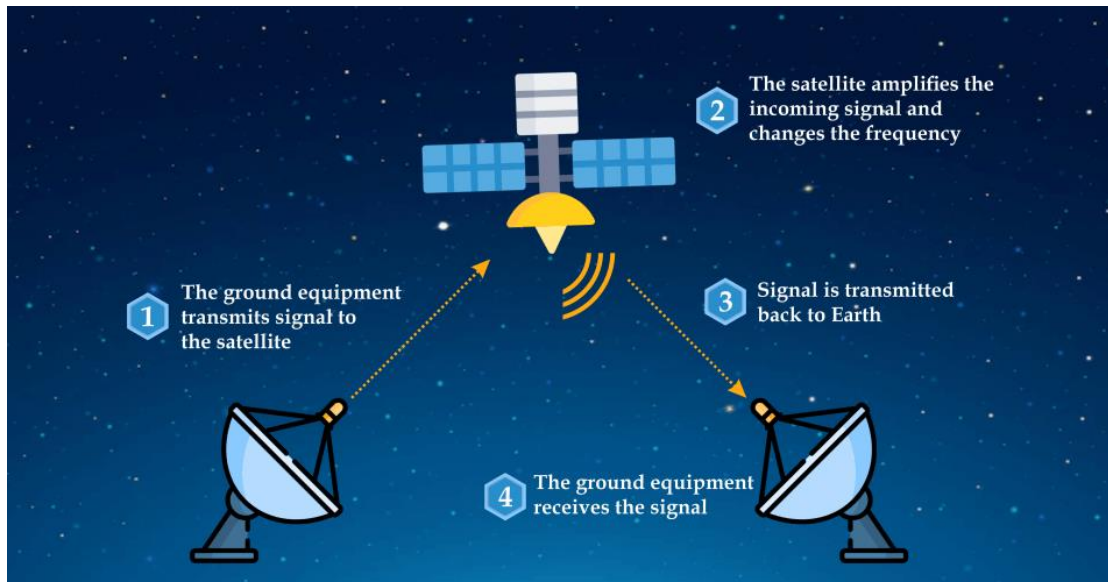


Figure 3. Satellite Communication [19].

1.7 Satellite frequency range

Satellite technology is rapidly expanding, fueled by advancements in the field. Satellites serve various purposes beyond radio communications, including applications in astronomy, weather forecasting, broadcasting, cartography, and many other areas [19]. The rapid expansion of satellite technology is driven by continuous advancements in engineering, miniaturization, and the increasing demand for global connectivity. Modern satellites are no longer limited to traditional communication roles; they now support a diverse range of applications that contribute significantly to scientific

research, environmental monitoring, and economic development. In astronomy, satellites such as space telescopes provide clearer and more detailed observations of distant celestial bodies, enabling groundbreaking discoveries about the universe. Weather forecasting has also seen significant improvements with the deployment of meteorological satellites, which collect real-time atmospheric data, track weather patterns, and predict natural disasters, aiding in disaster preparedness and response. Cartography has been transformed through high-resolution satellite imagery, facilitating accurate mapping and geospatial analysis for urban planning, natural resource management, and infrastructure development. Additionally, satellite-based remote sensing plays a crucial role in monitoring climate change, deforestation, and ocean health, contributing to sustainable development goals. The integration of satellite technology into broadcasting has revolutionized media consumption, allowing for global live streaming of news, sports, and entertainment events. As satellite technology continues to evolve, its scope of application is expected to expand further, supporting smart cities, autonomous vehicles, and advanced navigation systems, thereby shaping the future of global connectivity and technological progress.

1.7.1 L-band range (1–2 GHz)

The L-band is the frequency range between 1 and 2 gigahertz (GHz). This spectrum is commonly utilized in satellite communication, mobile networks, and various wireless applications. Its properties make it ideal for balancing the trade-off between signal propagation and data transmission. Its characteristics make it a perfect medium for information transmission and signal propagation trade-offs [19].

1.7.2 S-band range (2–4 GHz)

The S-band, spanning frequencies between 2 and 4 gigahertz (GHz), is a versatile range. It finds applications in radar systems, weather monitoring, and satellite communication, showcasing its adaptability across various fields. S-band signals, with their advantages in atmospheric absorption and penetration, are particularly suited for long-range radar applications. Furthermore, these frequencies are integral to mobile communication and aviation systems, further highlighting their versatility. The allocation of S-band frequencies is regulated by international telecommunications organizations, ensuring their widespread and efficient use, as shown in Figure 4 [18], [19]. Additionally, the S-band is integral to radar systems used in military defense and maritime navigation,

providing long-range detection of objects and ships even in adverse weather. The effective allocation and regulation of S-band frequencies by international telecommunications authorities ensure minimal interference and optimal usage across multiple sectors. As technology advances, the demand for S-band frequencies is expected to grow, particularly with the rise of 5G networks, unmanned aerial vehicles (UAVs), and smart transportation systems, which all require reliable, high-capacity communication channels.



Figure 4. Satellite Frequency Bands [19].

1.7.3 C-band range (4–8 GHz)

C-band frequencies, ranging from 4 to 8 gigahertz (GHz), are the workhorses of satellite communication, broadcasting, and radar systems. These frequencies, with their unique balance between atmospheric absorption and data transmission efficiency, are known for their remarkable resilience to rain fade. This feature ensures that satellite communication remains reliable even in adverse weather conditions. International telecommunications agreements regulate this spectrum, and its widespread use is a key driver of global connectivity through satellite links. The C-band is not just essential, it's a linchpin for various applications, including television broadcasting, telecommunication networks, and weather monitoring [20].

1.7.4 X-band range (8–12 GHz)

The X-band, encompassing frequencies from 8 to 12 gigahertz (GHz), is a crucial component in radar systems, satellite communication, and military applications. Its higher frequencies facilitate reliable and efficient satellite communication, while also offering benefits such as improved resolution in radar imaging and decreased atmospheric interference. In defense applications, it is frequently utilized for communication, surveillance, and missile guidance [21].

1.7.5 Ku-band range (12–18 GHz)

The Ku-band operates within the frequency range of 12 to 18 gigahertz (GHz) and is widely used for satellite communication, broadcasting, and broadband services. Its higher frequencies enable faster data transmission rates, making it suitable for high-speed internet via satellite. The Ku-band is particularly favored for its compatibility with smaller antennas, which enhances its use in consumer satellite services. However, it is susceptible to rain fade, though technological advancements have helped to address this issue. Additionally, the Ku-band is regulated internationally to ensure efficient and fair use across various communication services, including television broadcasting and broadband internet [21].

1.8 Wireless local area network (WLAN)

A Wireless Local Area Network (WLAN) offers the convenience of connecting devices to a local area network (LAN) without the hassle of physical wired connections. Using wireless protocols like Wi-Fi, WLANs typically cover limited areas such as homes, offices, or public spaces, making it easier for devices like computers, smartphones, and tablets to connect. Wi-Fi, operating on frequency bands like 2.4 GHz and 5 GHz, enables devices to connect through access points or routers. WLANs not only provide ease of mobility, allowing users to move around without losing connection, but also include robust security features like encryption and authentication to protect data, ensuring your information is safe and secure. WLANs, commonly used for internet access and file sharing, offer high-performance connectivity in various environments, including homes, offices, and cafes. This high-performance connectivity empowers users to work efficiently and stay connected. Additionally, BLE-enabled Aruba Tags enhance mobile engagement through applications for indoor wayfinding, asset tracking, and analytics. Wireless local area networks, often known as WLANs, are widely utilized for the purpose of gaining access to the internet and exchanging data. WLANs offer high-performance connectivity in a range of contexts, including homes, offices, and cafes. Wireless local area networks, often known as WLANs, are widely utilized for the purpose of gaining access to the internet and exchanging data. WLANs offer high-performance connectivity in a range of contexts, including homes, offices, and cafes. With the help of this high-performance connectivity, users are able to maintain their connections and work in an efficient manner. Enhanced mobile engagement with

analytics, asset tracking, and indoor navigation are all made possible by Aruba Tags that are equipped with Bluetooth Low Energy (BLE) connectivity [22]. Protection of data should be ensured by using effective security elements such as encryption and authentication, as shown in Figure 5.

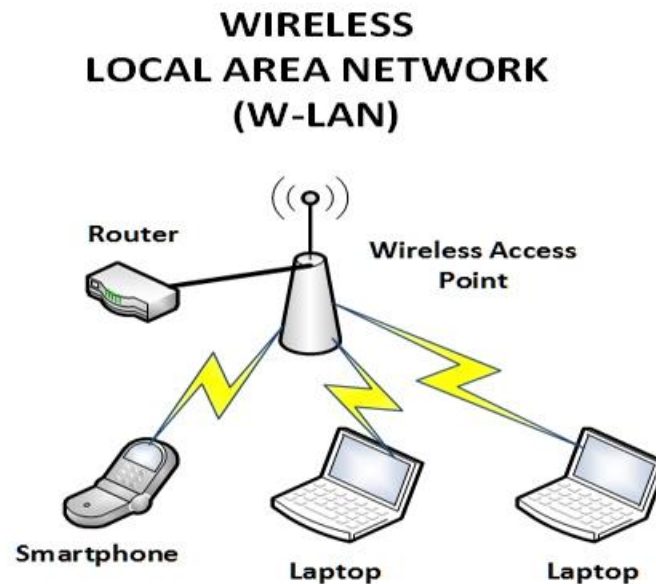


Figure 5. Wireless Local Area Network [23].

1.9 Weather radar systems

Weather radar systems are highly sophisticated tools that play an indispensable role in modern meteorology by detecting, monitoring, and analyzing various weather conditions. These systems utilize radio waves to detect precipitation, storm intensity, wind patterns, and other atmospheric phenomena. They are particularly important for early warning systems, providing vital data that helps meteorologists predict severe weather events like thunderstorms, hurricanes, tornadoes, and floods. The information gathered by weather radars allows meteorologists to issue timely alerts, thereby reducing the potential impact of extreme weather on communities and infrastructure. In addition to forecasting, weather radar systems are crucial for conducting scientific research that enhances our understanding of the Earth's atmosphere.

By continuously monitoring the atmosphere, these systems help researchers study the behavior of weather patterns, cloud formation, and atmospheric pressure systems. This research contributes to improving weather models and refining forecasting techniques, leading to more accurate predictions and better preparedness for natural disasters [25].

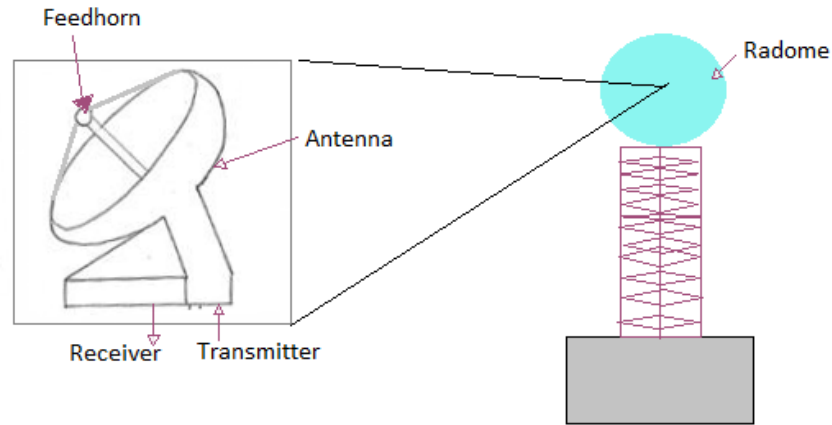


Figure 6. Weather Radar Systems [24].

As shown in Figure 6 The technology behind weather radar systems has evolved significantly over the years, with advanced systems now capable of providing detailed, real-time information on weather conditions. Doppler radar, for instance, allows meteorologists to measure the velocity of rain or wind particles, providing insights into wind direction, speed, and storm rotation. This capability is particularly useful for identifying the formation of tornadoes or other severe weather events. Furthermore, weather radar systems can operate in various frequency bands, allowing them to detect a wide range of atmospheric conditions, including light rainfall, snow, and even hail. As weather patterns become increasingly unpredictable due to climate change, weather radar systems are more important than ever in ensuring public safety and environmental protection. They provide the data necessary for proactive measures in disaster management, allowing authorities to make informed decisions about evacuations, road closures, and resource allocation during extreme weather events. Additionally, these systems are integral to aviation safety, helping pilots navigate through stormy weather and avoid hazardous conditions. In conclusion, weather radar systems are a cornerstone of modern meteorology, offering essential data for weather forecasting, research, and early warning systems. Their ability to monitor and analyze atmospheric conditions in real-time enhances our understanding of the Earth's weather patterns and supports efforts to mitigate the impact of severe weather events. As technology continues to advance, these systems will remain essential tools for ensuring the safety and well-being of communities worldwide. In conclusion, weather radar systems are a cornerstone of modern meteorology, offering essential data for weather forecasting, research, and early warning systems. Their ability to monitor and analyze atmospheric conditions in real-time enhances our understanding of the Earth's weather patterns and

supports efforts to mitigate the impact of severe weather events. As technology continues to advance, these systems will remain essential tools for ensuring the safety and well-being of communities worldwide.

1.10 Weather radar systems

"Radio Astronomy" by John D. Kraus, published in 1966 and updated in 1986, remains relevant today. It covers essential concepts in physics and electromagnetics that are still applicable. The book's practical focus, which is evident in its exploration of a variety of topics such as solar system radio sources, celestial radio sources, wave propagation, radio telescope antennas, and extraterrestrial intelligence (SETI), makes it a valuable resource for amateur and professional astronomers. The numerous illustrations and problems further enhance the hands-on learning experience [26].

Kraus's clear writing style and thorough explanations have established it as a classic in the field, often called the "bible of radio astronomy." It features 12 chapters and seven appendices, providing a comprehensive guide for those with a basic understanding of integral calculus. The Cygnus-Quasar edition further enhances its value with spiral binding and a solutions manual, ensuring its continued relevance despite technological advancements. Readers can be assured that they are getting a complete guide to radio astronomy [26].

1.11 Automotive radar systems

The future challenges for automotive radars are heavily shaped by two key trends that are transforming the industry and driving the need for innovation. The first trend is the integration of complex safety functions into small, high-volume car platforms. A prime example of this is the successful implementation of radar-based Collision Prevention Assist in the new Mercedes-Benz B-Class, which demonstrates the potential for compact radar systems to offer high levels of safety in affordable, mass-produced vehicles. This trend highlights the importance of miniaturizing radar technology while maintaining its performance, ensuring that sophisticated safety features become widely accessible to a larger population of drivers [26].

The second emerging trend is the necessity of deploying driver assistance and active safety systems across a wide variety of driving environments, from rural roads to busy

urban streets. This is where radar technology proves invaluable due to its superior ability to detect objects beyond the range of human perception. Unlike cameras or lidar, radar operates effectively in all weather conditions and lighting scenarios, making it the ideal sensor for systems that must function consistently in diverse and unpredictable environments.

However, these advancements and the growing demand for radar-based safety systems bring with them four significant challenges that developers must address to ensure continued success in the automotive radar market:

1. **Interoperability:** As radar systems are deployed globally, ensuring interoperability across various regions and vehicle models becomes crucial. Different countries and manufacturers may have different standards, so radar systems must be designed to operate seamlessly across all environments and vehicle platforms. Achieving this will require collaboration among international regulatory bodies and automakers to create universal standards for radar technology [26].
2. **Frequency Regulation:** One of the biggest hurdles facing radar developers is navigating global frequency regulations. The electromagnetic spectrum is a finite resource, and various countries have their own rules governing the use of frequencies, which can affect radar availability and performance. Developers need to ensure that their systems can function efficiently without interference from other devices that may be using the same or adjacent frequency bands [26].
3. **World Sensor Approach:** To meet the demands for mass adoption of radar technology in the automotive industry, manufacturers must focus on economies of scale. Developing radar systems that can be produced at a lower cost without sacrificing quality is essential for making advanced safety systems available to a broader range of vehicles. This approach requires innovation in radar sensor design, as well as efficient manufacturing techniques, to lower costs and increase accessibility. This approach requires innovation in radar sensor design, as well as efficient manufacturing techniques, to lower costs and increase accessibility. One of the biggest hurdles facing radar developers is navigating global frequency regulations. The electromagnetic spectrum is a finite resource, and various countries have their own rules governing the use of frequencies.

4. **Imaging Capabilities:** The growing complexity of urban driving environments, with their dense traffic and various obstacles, presents a significant challenge for radar systems. To effectively navigate these environments, radars need to incorporate advanced imaging capabilities that allow them to identify, track, and classify objects with greater precision. Developing radars with high-resolution imaging and enhanced object detection capabilities will be essential to ensuring that driver assistance systems can respond accurately in complex, real-world situations [27].

In addition to these challenges, there is a pressing need for radar sensors that can perform well on smaller car platforms, which are becoming increasingly popular in the automotive market. These smaller vehicles require radar systems that deliver robust standalone performance, ensuring that safety features like collision avoidance, lane-keeping assistance, and adaptive cruise control function reliably. At the same time, these systems must be cost-effective, striking a balance between affordability and advanced functionality. Achieving this will require creative solutions to integrate radar technology into compact sensor packages without compromising performance.

The automotive radar industry must tackle these challenges head-on to enable the next generation of vehicles equipped with state-of-the-art safety and driver assistance systems. By focusing on interoperability, regulatory compliance, cost reduction, and imaging advancements, radar technology will continue to evolve, ensuring that vehicles are safer, smarter, and more capable in all driving environments. As these innovations unfold, automotive radar will play a central role in shaping the future of transportation and driving safety [28].

CHAPTER 2

LITERATURE REVIEW

2.2 Background

The study of metamaterials has garnered significant attention due to their unique properties, particularly in the field of electromagnetic wave absorption. Metamaterials are artificially structured materials engineered to exhibit properties not found in naturally occurring materials [24]. They have applications in various domains, including stealth technology, solar energy harvesting, and advanced sensor systems. A dual-band metamaterial absorber (MMA) is specifically designed to absorb electromagnetic radiation at two distinct frequency bands, making it highly relevant for applications in C and X bands, which are critical for satellite communications and radar systems [25].

2.2.1 Categorization of metamaterials

Metamaterials can be categorized into two primary groups according to their mathematical representations. The first category comprises structures designated as Double Negative (DNG) and Single Negative (SNG), whilst the second group contains structures referred to as Photonic Band Gap (PBG) or materials displaying photonic bandgaps, used interchangeably, as shown in Figure 7.

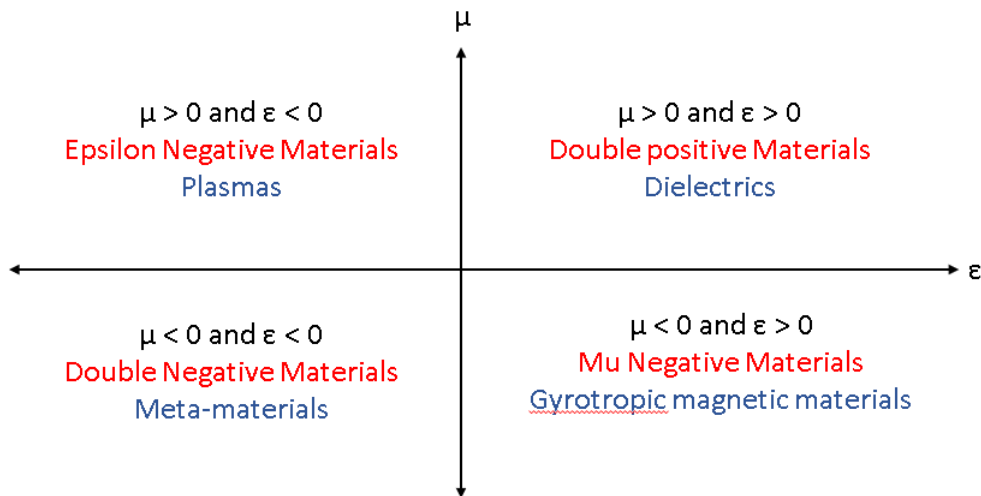


Figure 7. Categorization of materials based on their permittivity and permeability [25].

The primary group is designated as "double positive" (DPS) chemicals because they exhibit both ϵ and μ values exceeding zero. The majority of goods in this category are primarily dielectrics. In the second categorization, permittivity is negative while permeability is positive, resulting in its designation as an epsilon negative (ENG) material. This trait is frequently observed in various plasmas at particular frequencies. Materials classified in the third group possess a permittivity larger than zero and a permeability less than zero, categorizing them as mu-negative (MNG) materials, recognized for their gyrotropic magnetic characteristics. The fourth group consists of double negative (DNG) materials, which can alone be synthetically produced. This category of material demonstrates both negative permittivity and negative permeability. When an electromagnetic wave penetrates certain materials, it experiences modifications in its propagation properties. It is crucial to acknowledge that in nature, no materials exhibit both negative permittivity and negative permeability. From the aforementioned list, it can be inferred that metamaterials are a unique category of materials intentionally designed to exhibit negative permittivity and negative permeability [20]-[24]. Nonetheless, the continuous advancement of structures possessing distinctive features and applications has led to the formulation of a more extensive definition for designating materials as metamaterials. A metamaterial is a purposefully designed, large-scale composite with a periodic cellular structure that interacts with electromagnetic waves to achieve certain performance objectives not possible with naturally occurring materials [27].

2.2.2 Microwave metamaterial configuration

A Microwave Metamaterial Configuration denotes an artificially constructed structure designed at the tiny scale to manipulate and control electromagnetic radiation within microwave frequencies. This customized arrangement generally has distinct electromagnetic characteristics absent in naturally occurring substances. The composition and configuration of the metamaterial facilitate its interaction with microwaves in atypical manners, resulting in applications in communication, imaging, and sensing. The configuration utilizes the exact design of its components to produce customized reactions, such as negative refraction and effective permeability. These attributes enhance the efficacy of microwave devices and systems. The evolution of Microwave Metamaterial Configurations signifies a notable progress in materials science, presenting potential for the creation of innovative technologies with enhanced

capabilities in the microwave frequency spectrum. Researchers persist in investigating and enhancing these combinations for many practical applications within the domain of electromagnetic engineering [27].

2.2.3 Categorization system

Metamaterials can be classified into two primary types according to their mathematical representation. The first group includes structures identified as Double Negative (DNG) and Single Negative (SNG), whilst the second category comprises formations termed Photonic Band Gap (PBG) or materials acknowledged as photonic crystals. As previously mentioned, the internal inclusions within Double Negative (DNG) and Single Negative (SNG) materials are significantly smaller than the operating wavelength, leading to their overall homogeneity and characterisation through the concept of an effective medium. Conversely, the components of Photonic Band Gap (PBG) structures are situated at approximately half the wavelength or greater. Consequently, materials referred to as photonic crystals should not be regarded as homogeneous media. Bragg reflection is typically used for identification, a phenomena that minimally affects Double Negative (DNG) and Single Negative (SNG) structures, necessitating the use of different techniques for periodic media, as shown in Figure 8.

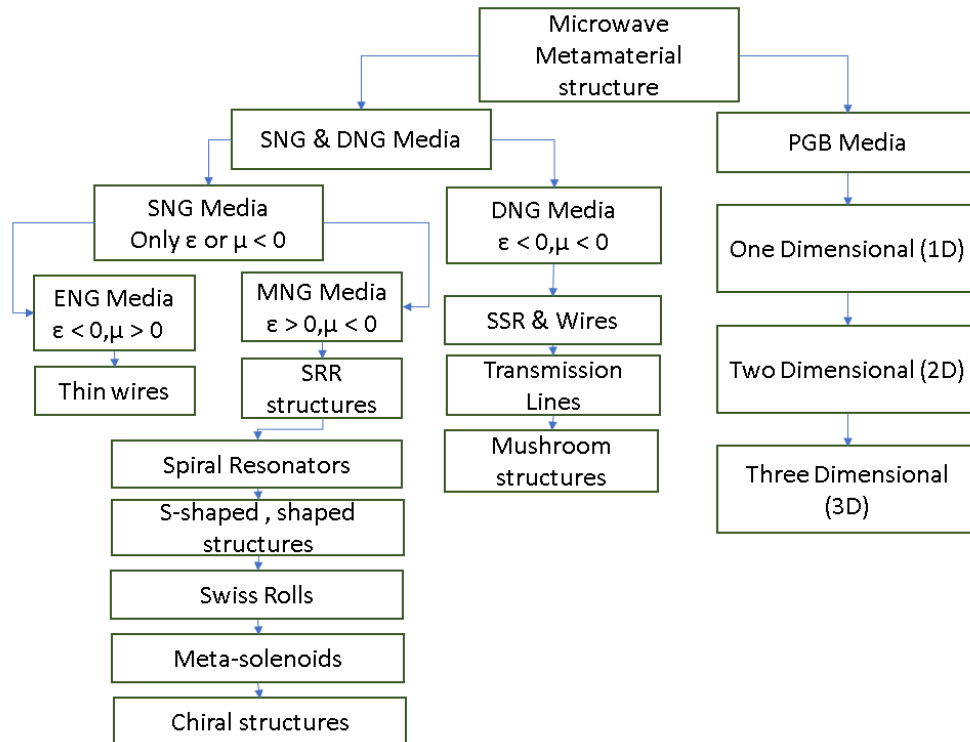


Figure 8. Categorization of Microwave Metamaterial Structures [27].

2.2.4 Negative epsilon metamaterials

Epsilon-negative (ENG) metamaterials are a distinct category of synthetic materials engineered to demonstrate negative permittivity (ϵ) at certain frequencies. In these materials, the electric polarizability is reversed, resulting in a contrary response to an applied electric field in comparison to natural materials.

This distinctive characteristic enables ENG metamaterials to engage with electromagnetic waves in atypical manners, resulting in phenomena like as negative refraction [28]. Negative permittivity is generally attained by arranging subwavelength features to form a composite material with customized electromagnetic characteristics. In the specified formula, ω_p represents the plasma frequency, while ω indicates the frequency of the propagating electromagnetic wave. The effective permittivity becomes negative when the frequency is below the plasma frequency, according to this equation. At the plasma frequency, the effective permittivity approaches zero, resulting in a refractive index of zero [29]. Epsilon-negative (ENG) metamaterials are synthetically created substances intended to display negative permittivity (ϵ) at particular frequencies. This distinctive attribute is critically articulated as $\epsilon(\omega) < 0$, where ω denotes the frequency of electromagnetic waves. Metamaterials attain negative permittivity by the configuration of subwavelength features. This unique feature facilitates atypical interactions with electromagnetic waves, resulting in phenomena such as negative refraction. The effective permittivity becomes negative when the frequency is below the plasma frequency (ω_p). At the plasma frequency, the effective permittivity equals zero, leading to a refractive index of zero. ENG metamaterials are utilized in optical and microwave engineering, facilitating the creation of advanced gadgets and functionality [27],[28].

2.2.5 Negative mu metamaterials

Mu-negative (MNG) metamaterials are synthetically engineered substances intended to demonstrate negative permeability (μ) at particular frequencies. The magnetic response in these materials is inverted, resulting in distinctive interactions with electromagnetic radiation. This feature is represented as $\mu(\omega) < 0$, where ω is the frequency of the electromagnetic wave. MNG metamaterials attain negative permeability by the configuration of subwavelength structures. This feature enables atypical electromagnetic phenomena, including negative refraction and backward-wave.

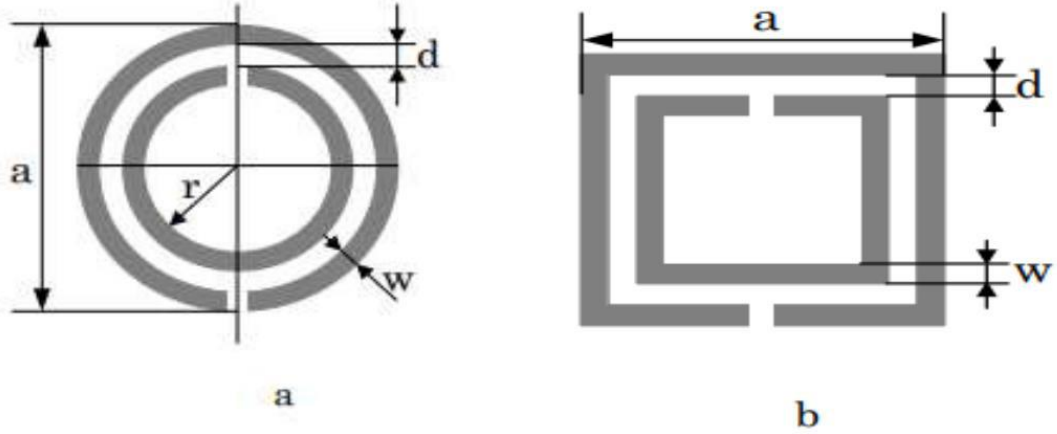


Figure 9. The initial unit cells of the mu-negative (MNG) material: a) Circular, b) Square [28].

elements to produce double-negative (DNG) metamaterials, facilitating enhanced manipulation of electromagnetic waves, as shown in Figure 9. Applications encompass antennas, imaging devices, and various microwave engineering technologies [28].

2.2.6 Metamaterial absorbers

Metamaterials are deliberately engineered structures exhibiting features that do not occur naturally, enabling them to influence electromagnetic (EM) waves. These materials generally have a dielectric substrate with metallic layers on one or both surfaces, exhibiting various patterns that include strips and capacitive gaps. Metamaterials can exhibit negative values for permittivity, permeability, and refractive index, affecting electromagnetic wave propagation within the dielectric medium of devices such as antennas, radar systems, or microwave sensors. The engineered metallic layers on the dielectric substrate significantly affect the overall permittivity and permeability, leading to the features of these synthetic metamaterials. Metamaterials are classified based on their demonstration of either a single negative (SNG) or double negative (DNG) value for permittivity and permeability in the dielectric substrate when subjected to electromagnetic wave action [28].

Generally constructed from thin layers of conductive and dielectric substances, it utilizes resonance phenomena to absorb and transform incoming radiation into thermal energy. This selective absorption renders it advantageous for applications including stealth technology, energy harvesting, and sensor systems. Metamaterial absorbers can their design facilitates enhanced absorption performance relative to traditional absorbers, hence advancing multiple technological domains [29].

2.2.7 Configuration of metamaterial absorber

A metamaterial absorber is a substance engineered to absorb electromagnetic waves at designated frequencies. It consists of synthetic structures, frequently designed at the nanoscale, to attain distinctive electromagnetic characteristics. The design of a metamaterial absorber is contingent upon the requisite absorption attributes, encompassing frequency range, bandwidth, and efficiency. Figure 2.4 illustrates a schematic illustration of an MMA [29].

MMA is engineered to permit electromagnetic (EM) waves to traverse it from any angle, aiming to efficiently absorb these EM waves at designated resonance frequencies, as shown in Figure 10. The construction includes unit cells designed to resonate at the desired frequency, enhancing absorption efficiency [29].

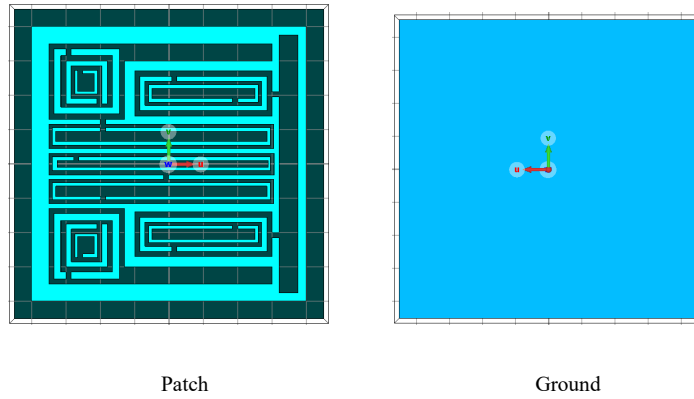


Figure 10. Visual Representation of Metamaterial Absorber Structure.

2.2.8 Characteristics of metamaterial absorber

The reflection coefficient, denoted as s_{11} , is a key parameter in evaluating how much of an incident electromagnetic wave is reflected back from a surface or material. It represents the ratio of the reflected wave to the incident wave at a specific interface, such as between a metamaterial surface and the incident electromagnetic field. Mathematically, s_{11} is a complex number that reflects both the magnitude and phase of the reflected signal. The square of the reflection coefficient, s_{11}^2 , is used to quantify the amount of energy reflected from the material. This value is important because it indicates how much of the wave does not pass through the material, giving insight into how reflective a surface or material is. The square of the reflection coefficient, s_{11} , is used to quantify the amount of energy reflected from the material. This value is important because it indicates how much of the wave does not pass through the material.

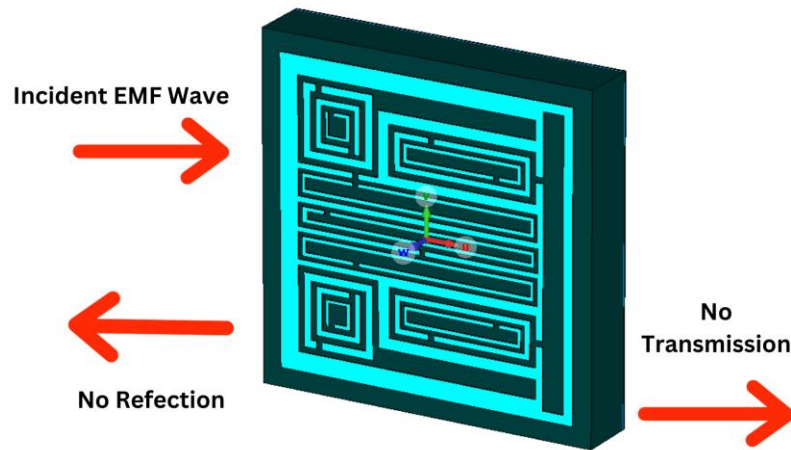


Figure 11. Absorption through a Metamaterial Absorber.

A pair of waveguide ports is typically employed at both ends of the Microwave Measurement Device (MMA) to evaluate its S parameters. The S parameters can be articulated in relation to the electric field, as delineated by Liu et al. in 2012. The electric field calculated at the port is a synthesis of the excitation and the reflected field. Metamaterial absorbers possess unique configurations referred to as unit cells. These unit cells are engineered to resonate at designated frequencies as shown in Figure 11. The configuration and measurements of unit cells dictate absorption properties. Metamaterial absorbers can be tailored for diverse purposes and frequency spectra. Precision production techniques, such as lithography, are utilized to produce nanostructures for enhanced performance [28],[29].

2.2.9 Single negative metamaterial absorbers

A Single Negative (SNG) Metamaterial Absorber is a customized metamaterial exhibiting unique electromagnetic properties. This falls within the group of metamaterials distinguished by negative permittivity and negative permeability, demonstrating distinctive reactions to incoming electromagnetic waves. The negative values of both properties facilitate concurrent negative refraction, rendering SNG metamaterials advantageous for applications such as lensing and cloaking devices. SNG metamaterial absorbers are generally designed using periodic unit cells that utilize resonant interactions to efficiently absorb specified frequencies. This resonant phenomenon results from the interplay of electric and magnetic resonances within the unit cells. SNG metamaterial absorbers can be designed to function within particular

frequency ranges, offering a degree of tunability and personalization. The applications of SNG metamaterial absorbers encompass several domains, including as telecommunications, medical imaging, and radar technology. The distinctive characteristics of SNG metamaterials enhance electromagnetic wave manipulation and offer potential for future developments in communication and sensing technology [27],[29].

2.2.10 Metamaterial absorber with near zero index

Metamaterials characterized by zero permittivity, zero permeability, or both at a particular frequency are referred to as zero-index metamaterials (ZIM). In these materials, electromagnetic waves exhibit quasi-infinite phase velocity and infinite wavelength, leading to an almost insignificant refractive index. This unique characteristic results in an almost uniform phase distribution, as the dipoles in these metamaterials oscillate in perfect synchrony. Thus, zero-index media enable the transmission of highly directed waves with negligible phase changes [25]. Zero-index metamaterials have the capacity to diminish the breadth of far-field antenna patterns, resulting in a radiation field with uniform phase, thereby producing an extraordinarily focussed and finely collimated outgoing beam across an extensive planar surface, as cited in [26]. Moreover, ZIM can enhance the directivity of microstrip patch antennas by proficiently controlling the emission direction, as outlined in [27]. The zero-index metamaterial category includes epsilon near zero (ENZ) and mu near zero (MNZ) metamaterials, with those exhibiting nearly zero real permittivity classified as epsilon-near-zero (ENZ) metamaterials.

Similarly, metamaterials with a real permeability value nearing 0 are classified as Mu near-zero (MNZ) metamaterials. These materials possess prospective applications in spatial filtering [28], improving radiation directivity, and facilitating the efficient coupling and concentration of electromagnetic energy [28]. Utilizing their swift wave propagation characteristics and inherent negative polarizability, ENZ metamaterials can be employed in transparency and cloaking applications [29].

2.2.11 Previous work review

A comprehensive review of previous works reveals a variety of designs and methodologies employed in the development of dual-band metamaterial absorbers:

- A study introduced the concept of a perfect metamaterial absorber, laying the groundwork for subsequent research in this area. Their work demonstrated the potential for high absorption rates across a range of frequencies [28].
- Another study presented a filling-factor engineered MMA capable of operating effectively at multiple frequencies within the C and X bands. Their findings indicated an average absorption rate exceeding 99%, showcasing advancements in material engineering and design strategies [29].
- Dual-band absorbers with polarization insensitivity have been explored, offering flexibility in application scenarios. Researchers have investigated dual-band absorbers that are insensitive to polarization, which opens up a range of potential uses.
- Lightweight and ultra-wideband absorbers utilizing multilayer dielectric structures have also been developed to address size and bandwidth limitations.

2.3 Performance analysis

The selection of journal papers for this literature review was guided by criteria including relevance to dual-band metamaterial absorbers, publication in reputable journals, and contributions to advancements in design and application strategies. Key journals considered include:

- Journal of Materials Research and Technology
- International Journal of Microwave and Wireless Technologies
- Journal of Electromagnetic Waves and Applications

The design of metamaterial absorbers, a field of constant innovation, has been extensively explored in numerous books and published articles. Achieving the desired absorption properties often involves experimenting with various geometries, such as resonant structures, frequency-selective surfaces (FSS), and gradient-index materials.

To enhance the performance of these absorbers, researchers frequently employ theoretical analyses, simulations, and optimization techniques. A common and significant goal in the study of metamaterial absorbers is to achieve broadband absorption, which allows a wide range of frequencies to be absorbed. This is crucial as

it expands the potential applications of these absorbers. Various designs and structures proposed in the literature aim to broaden the absorption bandwidth by optimizing geometric factors, material properties, and resonant structures. Another critical aspect of metamaterial absorber design is maximizing absorption efficiency. Many studies aim to improve the absorber's ability to collect and convert incident electromagnetic waves into heat or other valuable forms of energy. To enhance this efficiency, strategies such as impedance matching, multiple resonances, and polarization selectivity have been explored. Recent research indicates that metamaterial absorbers can perform functions beyond simple absorption. Some studies investigate tuneable absorbers with features like frequency tunability or polarization selectivity. Other research has focused on absorbers with additional capabilities, such as wavefront modification (e.g., shaping the wavefront of incident radiation), sensing abilities (e.g., detecting changes in the environment), or integration with other devices (e.g., combining with antennas for improved performance). A recurring theme in publications on metamaterial absorbers is the fabrication process for practical applications. These articles describe various fabrication methods, including lithography, nanoimprint lithography, additive manufacturing, and etching techniques. The emphasis is on developing scalable, cost-effective, and reliable manufacturing processes to produce high-performance metamaterial absorbers. Many of these studies also include experiments to validate the results of metamaterial absorbers. This often involves creating prototypes and measuring how well practical outcomes match theoretical predictions. The rigorous process of experimental validation supports the proposed designs and provides concrete evidence of the absorbers' effectiveness, instilling confidence in the research. Metamaterial absorbers, with their potential to revolutionize various applications, are being explored for communication, energy harvesting, sensing, imaging, and stealth technology. Publications often highlight the advantages, challenges, and potential applications of metamaterial absorbers across different fields while also speculating on their future impact. As research on metamaterial absorbers continues to evolve, it is advisable to consult current scientific databases, specialized journals, and conference proceedings focused on electromagnetic metamaterials and related areas to gain a comprehensive and up-to-date understanding of their performance and specific works. This strategic selection ensures a robust foundation for analyzing current trends and identifying gaps in existing research. The emphasis is on developing scalable, cost-effective, and reliable manufacturing processes to produce high-performance.

2.4 Recent paper analysis

The analysis of recent papers in the field of dual-band metamaterial absorbers (MMAs) reveals significant advancements and diverse approaches aimed at optimizing performance for C and X band applications. This section synthesizes findings from various studies, highlighting their contributions to the design, functionality, and application of MMAs.

Research on dual-band metamaterial absorbers (MMAs) has recently seen a plethora of new developments and methods for improving their performance in both the C and X bands. The results of different investigations are summarized below, with an emphasis on how these studies improved the design, operation, and use of MMAs.

2.5 Overview of recent studies

A Study introduced the concept of a perfect metamaterial absorber, demonstrating the potential for high absorption rates across a range of frequencies. The study established critical parameters for designing MMAs, including resonant structures and material selection, which have influenced subsequent research in this area [28].

An additional inquiry was carried out with the intention of developing a multi-frequency amplifier (MMA) that is capable of operating across a wide range of frequencies. This was done in order to fulfill the requirements that were set forth by the IEEE for the C and X bands. The creation of this MMA was also made possible through the utilization of filling components. In light of the fact that the design was successful in achieving an average absorption rate of 99% [29], it is abundantly obvious that there is a need to enhance the geometric arrangements in order to increase the efficiency of absorption as shown in Figure 12. Further research was undertaken to design a Multi-Frequency Amplifier (MFA) capable of operating seamlessly across a broad spectrum of frequencies, with a particular focus on meeting the IEEE standards for C and X bands. The development process involved the strategic use of filling components, which played a crucial role in optimizing the amplifier's performance. The successful design of this MFA achieved an impressive average absorption rate of 99%, demonstrating its high efficiency and effectiveness in frequency absorption. This achievement highlights the potential for further enhancements in the geometric configuration of the amplifier to improve its absorption efficiency. By refining the

arrangement and structure of the filling components, it is possible to achieve better energy capture, minimize signal loss, and support a wider range of frequency applications. Such improvements are critical for advancing radar, satellite communication, and wireless transmission technologies, where efficient absorption and signal clarity are paramount. The remarkable absorption rate of the MFA underscores the effectiveness of its design and suggests promising future developments in multi-band amplifier technology, as illustrated in Figure 12.

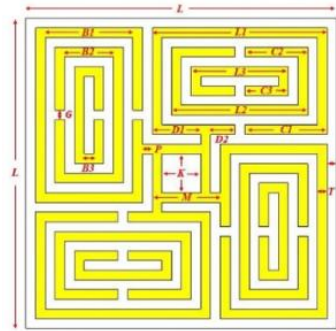


Figure 12. Referred Absorber [29].

For instance, a study presented a quad-band ultrathin MMA utilizing four resonators, achieving notable absorption across multiple frequency bands, which is crucial for applications requiring versatility in electromagnetic wave management as shown in Figure 13 [30].

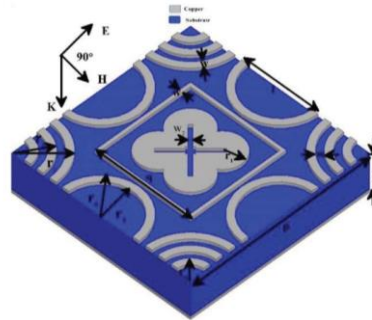


Figure 13. Appointed Recipient [30].

Another study has showed Another significant advancement is the development of lightweight, ultra-wideband MMAs using multilayer dielectric structures as shown in Figure 14. A paper highlighted designs that effectively operate over a wide frequency range (3.7–13.6 GHz), which is essential for commercial and military applications in C and X bands [31]. Another significant advancement in the field of multi-frequency amplifiers (MFAs) is the development of lightweight, ultra-wideband MMAs utilizing

multilayer dielectric structures, as illustrated in Figure 14. This innovative approach has enabled the creation of amplifiers that can effectively operate over an extensive frequency range of 3.7–13.6 GHz, covering both the C and X bands. Such a broad operational range is essential for meeting the demanding requirements of commercial and military applications, including radar systems, satellite communications, and electronic warfare. The use of multilayer dielectric structures not only enhances frequency coverage but also contributes to the reduction of overall weight and size, making the amplifiers more suitable for mobile and space-constrained applications. A 2021 study demonstrated the efficacy of this design, showcasing significant improvements in performance, energy absorption, and signal clarity. The reduced weight of the amplifier also facilitates its integration into modern, compact communication and defence systems, supporting next-generation technological advancements. As the demand for high-frequency, wideband communication continues to rise, such innovative designs are expected to play a crucial role in advancing the capabilities of radar, satellite, and military communication systems.

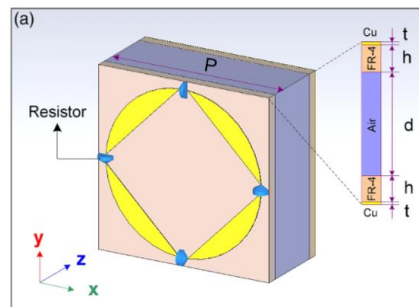


Figure 14. Nominated Absorber [31].

Another study, introduced an angle-independent wideband MMA, showcasing the ability to maintain high absorption efficiency across varying angles of incidence. This feature is particularly beneficial for real-world applications where electromagnetic waves may not always strike the absorber perpendicularly as shown in Figure 15 [32].

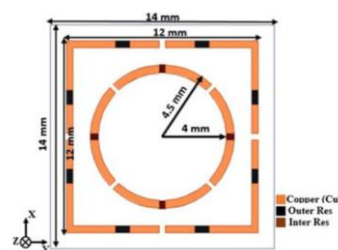


Figure 15. Designed absorber [32].

Polarization Independent Triple Band Ultrathin Conformal Metamaterial Absorber for C- and X-Frequency Bands, published in the International Journal of Electronics and Communications, as shown in Figure 16 this study presents a novel ultrathin conformal metamaterial absorber designed to operate effectively across three frequency bands, including the C and X bands. The authors emphasize the importance of polarization independence, which allows the absorber to maintain high performance regardless of the angle of incident electromagnetic waves. The design utilizes a conformal structure that enhances adaptability in various applications, particularly in environments where space is limited [33].

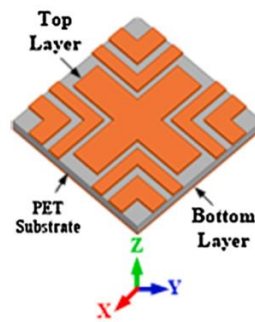


Figure 16. Nominated Absorber [33].

Tree-Shaped Fractal Metamaterial Multiband Absorber for C- and X-Band Applications [33], featured in the Journal of Electromagnetic Waves and Applications, introduces a tree-shaped fractal design for a multiband absorber tailored for C and X band frequencies. The fractal geometry contributes to a compact design while achieving effective absorption across multiple bands. The study highlights the advantages of using fractal structures in metamaterials, such as improved bandwidth and absorption efficiency, making it suitable for applications in telecommunications and radar systems.

2.6 Parameter selection

The selection of the right parameters is critical, as they can greatly influence both the outcomes and the area of application. The decision regarding these parameters is fundamental to the efficacy and performance of a metamaterial absorber, making them crucial in the development phase. Metamaterial absorbers, which are the devices you're focusing on, are specifically engineered to effectively capture electromagnetic waves, usually within a targeted frequency spectrum. The design parameters utilized establish important characteristics of a metamaterial absorber, such as absorption bandwidth, the

frequency at which peak absorption occurs, and sensitivity to polarization. As engineers and researchers in materials science and electromagnetic applications, we understand that our role in selecting the dimensions, shape, and arrangement of metamaterial unit cells is vital [33]. These parameters significantly impact the absorption efficiency of metamaterial absorbers, making our expertise essential for maximizing the absorption of incoming electromagnetic waves within the desired frequency range. Our choices regarding these parameters directly influence the bandwidth and frequency selectivity of the metamaterial absorber. Depending on the specific requirements of our applications, we can design absorbers with either broad bandwidth or narrowband absorption, which considerably affects the overall performance of the absorber. Metamaterial absorbers can be classified into two types: polarization-sensitive and polarization-insensitive [33]. The configuration and arrangement of the unit cells that we meticulously design determine the absorber's response to different incident wave polarizations. Our design decisions can yield polarization-insensitive absorbers for applications that do not require polarization filtering or polarization-sensitive absorbers for scenarios where such filtering is necessary. When selecting parameters for the design of a metamaterial absorber, we must consider the limitations imposed by fabrication and manufacturing techniques. Practical constraints, such as the resolution of the production process and material availability, must be carefully evaluated. Moreover, we are aware that parameter variations affect the number of capacitive and inductive elements, which in turn influences the resonance frequency. The resonance frequency is the point at which the absorber is most effective in absorbing electromagnetic waves. The number of capacitive and inductive elements also influences the peak resonance frequency and position, among other characteristics. Therefore, we need to maintain a mathematically correct ratio between the capacitive and inductive elements, requiring adjustments to the dimensions of the patches accordingly. As an engineer and researchers in materials science and electromagnetic applications, we understand that our role in selecting the dimensions, shape, and arrangement of metamaterial unit cells is vital. These parameters significantly impact the absorption efficiency of metamaterial absorbers, making our expertise essential for maximizing the absorption of incoming electromagnetic waves within the desired frequency range. Our choices regarding these parameters directly influence the bandwidth and frequency selectivity of the metamaterial absorber. Depending on the specific requirements of our applications, we can design absorbers with either broad

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Selecting appropriate parameters is crucial for the effective design of dual-band MMAs:

- **Resonant Frequency:** Targeting specific resonant frequencies within the C and X bands is essential for maximizing absorption efficiency.
- **Material Properties:** The choice of materials (e.g., copper for patches) significantly influences performance characteristics such as conductivity and dielectric properties.

2.7 Material selection

Choosing the appropriate materials is essential when building a metamaterial absorber, as it directly affects the electromagnetic properties and overall performance of the absorber. The material selection influences important electromagnetic attributes, such as complex permittivity and permeability, which determine how absorption, reflection, and transmission behave with respect to electromagnetic waves. To optimize absorption performance within a designated frequency range, it is vital to select materials that have

specific electromagnetic properties. The materials used impact the effectiveness and bandwidth of absorption in the metamaterial absorber, as different materials exhibit varying absorption characteristics. For example, metals with high electrical conductivity are frequently employed to achieve wide absorption bandwidths, while high-efficiency narrowband absorbers can be created using dielectric materials. Materials chosen for metamaterial absorbers should possess strong damping characteristics and minimal losses. Low-loss materials improve absorption efficiency by reducing the reflection and transmission of electromagnetic waves. Furthermore, materials with strong damping capabilities assist in dissipating the absorbed energy as heat, resulting in effective absorption. In certain uses, metamaterial absorbers may need to endure mechanical stress or extreme environmental situations. Therefore, aspects such as mechanical strength, durability, thermal stability, and resistance to moisture and chemicals should be taken into account when choosing materials. It is also necessary to consider the cost and availability of these materials. Depending on the needs of the project, selecting materials that are cost-effective and easily obtainable in adequate quantities could be critical. Material selection plays a pivotal role in the performance of metamaterial absorbers [31]-[34]:

- **Conductive Materials:** Metals like copper are commonly used due to their excellent conductivity.
- **Substrate Materials:** FR-4 substrates are frequently utilized for their balance between cost-effectiveness and performance characteristics.

In conclusion, this literature review synthesizes key findings from previous research while establishing a framework for further investigation into dual-band metamaterial absorbers tailored for C and X band applications. The insights gained will inform the design and simulation phases of this thesis project, ultimately contributing to advancements in this cutting-edge field [33]-[34].

CHAPTER 3

METHODOLOGY

3.1 Methodology

The methodology for studying metamaterial absorbers is an iterative process that involves setting research objectives, carrying out a materials analysis, designing the absorber shape, assessing performance, refining the design through a literature review, choosing suitable fabrication methods, analyzing the structure's effectiveness, executing simulations and experiments, conducting a parameter sensitivity analysis, exploring application and integration possibilities, recording findings for publication, and considering the necessity for further research. This iterative scientific approach enables the creation of materials that effectively capture electromagnetic waves and have the potential to be applied across various fields of research. It enhances our comprehension of metamaterial absorber technology and promotes its further advancement [29].

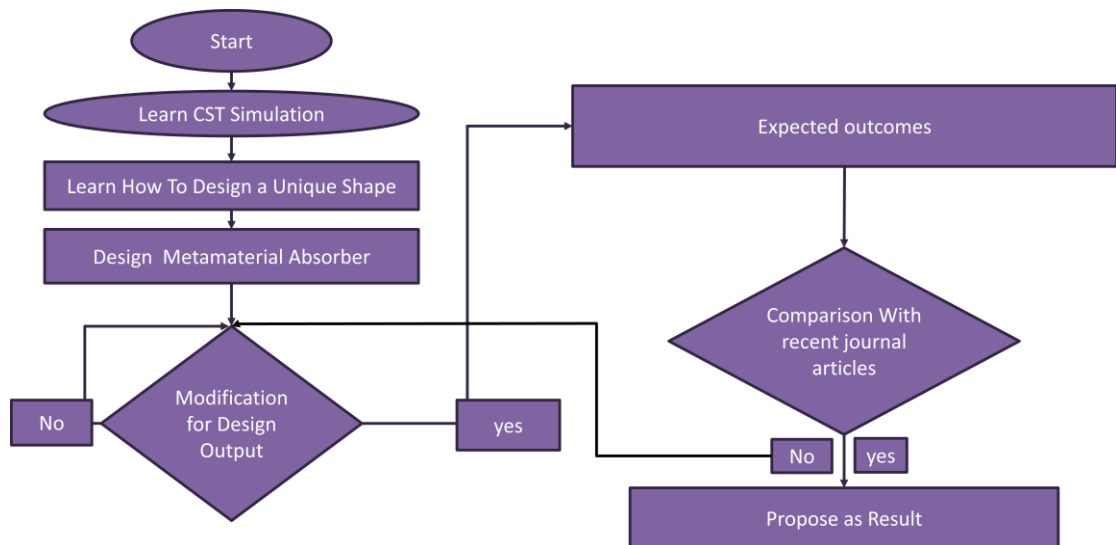


Figure 17. Methodology to design a metamaterial absorber.

The study of metamaterial absorbers can be divided into several distinct but interrelated steps. The first phase involves defining the target frequency range, efficiency, and bandwidth for absorption. Following this, a thorough review of existing materials is conducted to assess current understanding and pinpoint any gaps in knowledge. When choosing materials, as shown in Figure 17 it is it's crucial to consider both the

anticipated operating frequency and the desired characteristics. Next, the absorber's unit cell design is created, often utilizing simulation software to optimize various configurations. Fabrication methods such as lithography and 3D printing produce the actual physical structure. Characterization of the absorber is performed through experiments, usually involving the measurement of reflection and transmission. These results help evaluate the performance of the absorber. Equally important is the continuous refinement of design and manufacturing processes, which leads to the best possible outcomes and is validated using simulations and models [30]. Sensitivity analysis is used to examine how changes in specific parameters affect outcomes. The potential for applications and integration is also explored, and the findings are carefully documented for future publication. Prospective future research directions and challenges are also considered to advance the field of metamaterial absorbers. This systematic approach, which combines theoretical, computational, and experimental elements, is a testament to the collaborative nature of our field. It is through this shared knowledge and community effort that we develop materials with specific absorption properties for various applications.

Emphasize the Significance of the Study's Objectives: Clearly outlining the research goals is not just essential, but it also sets the direction for the study. It's crucial to identify the specific wavelengths or frequencies at which the metamaterial absorber should demonstrate its effectiveness, as well as the optimal bandwidth and absorption efficiency to be achieved [31].

Stress the Importance of a Comprehensive Literature Review: To gain insights into the industry's current state, prevailing models, and recent advancements in metamaterial absorbers, it's crucial to undertake an extensive review of existing publications. This approach not only helps identify knowledge gaps but also highlights potential topics for further investigation, providing a solid foundation for the research [32].

Material selection: Ensure that appropriate materials are chosen for designing the metamaterial absorber. The performance of these metamaterials, which typically consist of periodic structures, can be significantly influenced by the selection of materials used in their fabrication [34]. Ensure that appropriate materials are chosen for designing the metamaterial absorber. Prospective future research directions and challenges are also considered to advance the field of metamaterial absorbers of This systematic approach.

Leverage Simulation Software for Design and Structure: It's crucial to determine the configuration of the metamaterial unit cell and its design specifications. The unit cells must have the correct dimensions and shapes to absorb the target frequencies effectively. Utilizing simulation software such as HFSS, COMSOL, or CST Microwave Studio is not just beneficial, but it's also essential in generating an optimized design, aiding in the efficient and effective design of the metamaterial absorber [35].

Creating the desired effect: When designing the metamaterial absorber, various design parameters must be considered. Standard manufacturing techniques include photolithography, electron beam lithography, and 3D printing, with the choice of method often depending on production scale and frequency [35].

Characterization of the absorber: Before evaluating the performance of the fabricated metamaterial absorber, specific attributes must be defined. Reflection and transmission spectroscopy can measure critical characteristics, including absorption efficiency, bandwidth, and other relevant factors [35].

Focus on optimization: Adjusting design parameters and manufacturing processes can enhance the metamaterial absorber's performance. It's important to remember that achieving the desired absorption characteristics may require several iterations of the optimization process. This iterative approach will test your patience and persistence, but it will ultimately lead to a more robust and effective absorber [35].

Simulated and modeled systems: Utilize electromagnetic modeling and simulation tools to analyze and verify the behavior of the metamaterial absorber. This will provide you with a high level of confidence and reassurance as you compare simulation results with data obtained from experimental testing, ensuring the consistency and reliability of your findings [36].

Unveiling the delicate relationship among the parameters: Our investigation into how changes to various design parameters impact the absorber's performance may reveal not only essential components but also potential trade-offs, adding a layer of complexity and depth to our research [32].

Exploring the potential of metamaterial absorbers: Delve into metamaterial absorbers' versatility and potential in various applications, including imaging, energy harvesting, and stealth technology. Prospective future research directions and

challenges are also considered to advance the field of metamaterial absorbers. This systematic approach, which combines theoretical, computational, and experimental elements, is a testament to the collaborative nature of our field. It is through this shared knowledge and community effort that we develop materials with specific absorption properties for various applications.

E-publishing industry and the archiving sector: We must meticulously document our research's findings and conclusions. This will allow us to prepare research articles for submission to relevant academic journals and present our findings at conferences, advancing scientific knowledge. It is important to share our results with fellow researchers, fostering collaboration and the exchange of ideas.

Actual responsibilities and tasks: We discuss various ways metamaterial absorber technology could be enhanced and the different research domains that could be explored in the future. We identify any challenges or unresolved questions that need further investigation [34].

It is important to work alongside specialists in electromagnetic theory, materials science, and engineering throughout the research process. This teamwork will provide valuable knowledge and experience regarding the design and performance evaluation of metamaterial absorbers. Engaging with these experts allows for a deeper understanding of all the factors involved in creating and assessing metamaterial absorbers [32]-[35].

3.2 Design of Metamaterial Absorbers

Figure 1 is an illustration of the design of the metamaterial absorber that has been proposed. This absorber made of metamaterial is composed of three layers respectively. The middle of this individual unit cell is made up of a dielectric substance, which serves to separate the upper metal patch from the lower metal plane. This is accomplished by employing a FR-4 material that is 1.6 millimeters thick and has a relative permittivity (ϵ_r) of 4.3. The dimensions of the substrate layer are 20 millimeters by 20 millimeters. Copper, with a conductivity (σ) of $5.8 \times 10^7/\text{m}$ and a thickness of 0.035mm, is used to construct both the inner and exterior surfaces of the object. As part of the process of developing the metamaterial, we utilized a three-dimensional coordinate system that was contained within the orientation program known as CST Microwave Studio. The

patch also makes use of copper, with the FR-4 layer measuring 1.6 millimeters in thickness and the copper layer at 0.035 millimeters in thickness. The following table

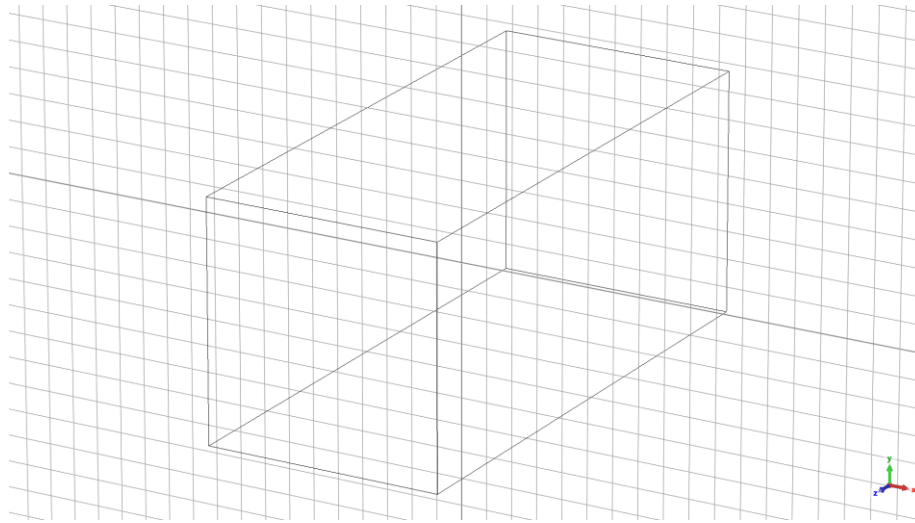


Figure 18. 3D co-ordination view in CST software.

provides an outline of particular geometric parameters. The CST STUDIO SUITE, which is a commercial full-wave modeling software, is utilized in order to construct and evaluate the unit cell of the metamaterial absorber as shown in Figure 18.

3.3 Substrate-layer

The width and length of the substrate are both determined to be 10 millimeters, and its thickness is 1.6 millimeters. The substrate material that has been utilized in this design is FR-4, and it has been used without any losses as shown in Figure 19. The calculated dimensions of the substrate are 10 mm in width, 1.6 mm in length, and 1.6 mm in thick.

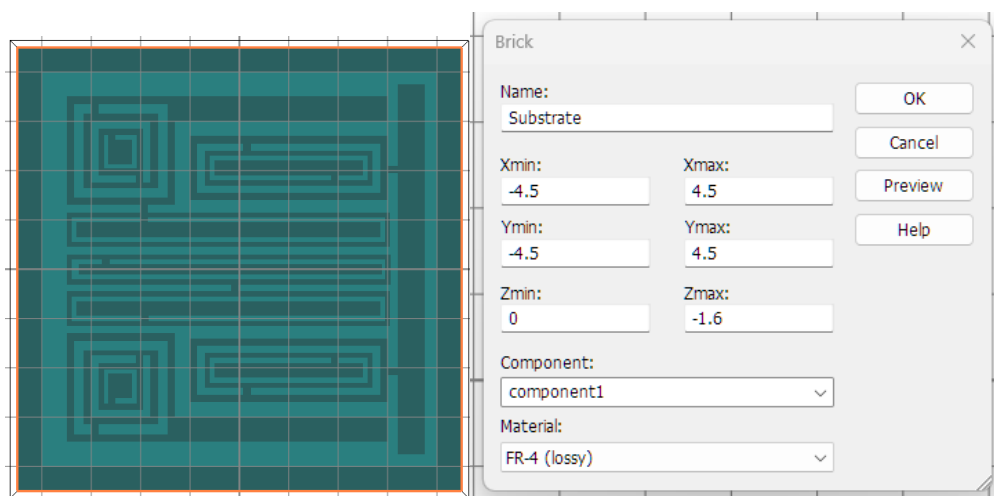


Figure 19. Substrate-layer.

The calculated dimensions of the substrate are 10 mm in width, 1.6 mm in length, and 1.6 mm in thickness. There have been zero losses in the usage of FR-4 as the substrate material in this design.

3.4 Patch Layer

Figure 3.4 illustrates the final layout of the metamaterial absorber (MMA) patch, showcasing its comprehensive design and structure. The patch layout is the culmination of several design iterations aimed at optimizing the sensor's performance in terms of resonance, sensitivity, and electromagnetic wave absorption. The layout reflects a systematic arrangement of unit cells, each configured with precision to achieve the desired absorption characteristics. The structure is meticulously aligned to ensure uniformity in electromagnetic response across the surface as shown in Figure 20.

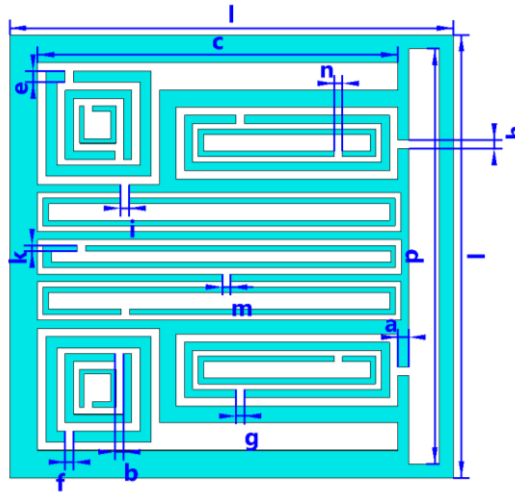


Figure 20. Patch Layer.

The integration of the layout into a single line highlights the innovative approach used to streamline the sensor's design. This integration enhances the overall compactness and efficiency of the device, reducing its physical footprint while maintaining high sensitivity and accuracy. By incorporating the entire layout into a single line, the sensor's design achieves a balance between simplicity and functionality, enabling it to be easily embedded into flexible, wearable, or portable devices. This single-line configuration also facilitates the seamless transfer of electromagnetic signals, minimizing signal loss and optimizing the resonance response. The continuous path created by the single line ensures uninterrupted current flow, thereby improving the sensor's responsiveness to changes in the surrounding dielectric environment. This

feature is particularly advantageous for applications in wearable sensors, where compact, lightweight, and flexible designs are essential.

TABLE I. ENUMERATION OF STRUCTURAL CHARACTERISTICS RELEVANT TO THE SUGGESTED MMA.

Parameters	a	b	c	d	e	f	g	h	i	k	l	m	p
Values(mm)	0.2	0.2	6.5	7.5	0.2	0.2	0.2	0.2	0.2	0.1	8.0	0.1	0.2

The table, labeled as TABLE I, provides a detailed enumeration of the structural characteristics relevant to the design and implementation of the suggested metamaterial absorber (MMA). Each parameter listed corresponds to specific dimensions within the MMA layout, all of which are critical to achieving the desired electromagnetic properties, such as optimal absorption and resonance performance.

The table outlines a total of 14 parameters (a through p), with their respective dimensions specified in millimeters (mm). These parameters collectively define the geometry of the MMA structure, including the lengths, widths, spacings, and thicknesses of various components within the patch. Each parameter contributes to the overall functionality and precision of the design, ensuring that the absorber operates efficiently within the intended frequency range.

- **Parameters a, b, e, f, g, h, i, m, and p (0.1–0.2 mm):** These represent fine structural details such as gaps, edges, and spacings. Their small dimensions are indicative of the high precision required to tune the resonator's electromagnetic response. These features play a crucial role in achieving resonance at specific frequencies and improving absorption performance.
- **Parameters c and d (6.5 mm and 7.5 mm):** These likely define the overall size or major dimensions of the patch, such as the length and width of the resonator or the unit cell. These dimensions are directly correlated with the operating wavelength and frequency, as they determine the metamaterial's ability to resonate with incoming electromagnetic waves.
- **Parameters k and l (8.0 mm and 0.1 mm):** These may define larger structural features or outer boundaries of the MMA layout. Parameter l, being relatively thin, might represent a critical boundary or layer thickness, while parameter k. This detailed enumeration allows for precise fabrication of the MMA design, as even minute deviations in these dimensions can significantly affect the

absorber's performance. The small gaps and spacings ensure strong electromagnetic coupling, while the larger dimensions control the overall resonance behavior.

By meticulously defining these parameters, the MMA design achieves optimal absorption, enhanced sensitivity, and reliability in its intended application, such as sensing, energy harvesting, or wave filtering. The table serves as a reference for manufacturing and simulation, ensuring that the design specifications are met for consistent performance across different implementations.

3.5 Incorporate a waveguide port

When setting up the waveguide ports for the simulation of a metamaterial absorber in CST Studio Suite, it is essential to properly configure the ports to ensure accurate and efficient simulation results. The purpose of using two waveguide ports—labeled as positive and negative—is to control and monitor the electromagnetic wave's propagation through the metamaterial. The waveguide ports act as the entry and exit points for the electromagnetic waves, facilitating the measurement of reflection and transmission coefficients (S_{11} and S_{21}).

The positive port is typically placed at the front or the source side of the metamaterial. This port introduces an electromagnetic wave into the system, simulating the incoming wave energy that interacts with the metamaterial structure. On the other hand, the negative port is positioned at the opposite side of the metamaterial, where the transmitted wave exits. This arrangement allows for the calculation of how much of the incoming wave is absorbed, reflected, or transmitted through the structure. To establish the correct flow of the electromagnetic wave, the ports must be aligned in such a way that the direction of propagation is consistent. In CST, this is achieved by setting the direction of the electric field (E-field) and magnetic field (H-field) relative to the port's position.

The electric field is typically perpendicular to the port surface, while the magnetic field runs parallel. The wave vector (k) points in the direction of wave propagation, flowing from the positive port to the negative port. This arrangement ensures that the wave moves in a single, well-defined direction, simulating real-world conditions for devices like absorbers and sensors. In CST, this is achieved by setting the direction.

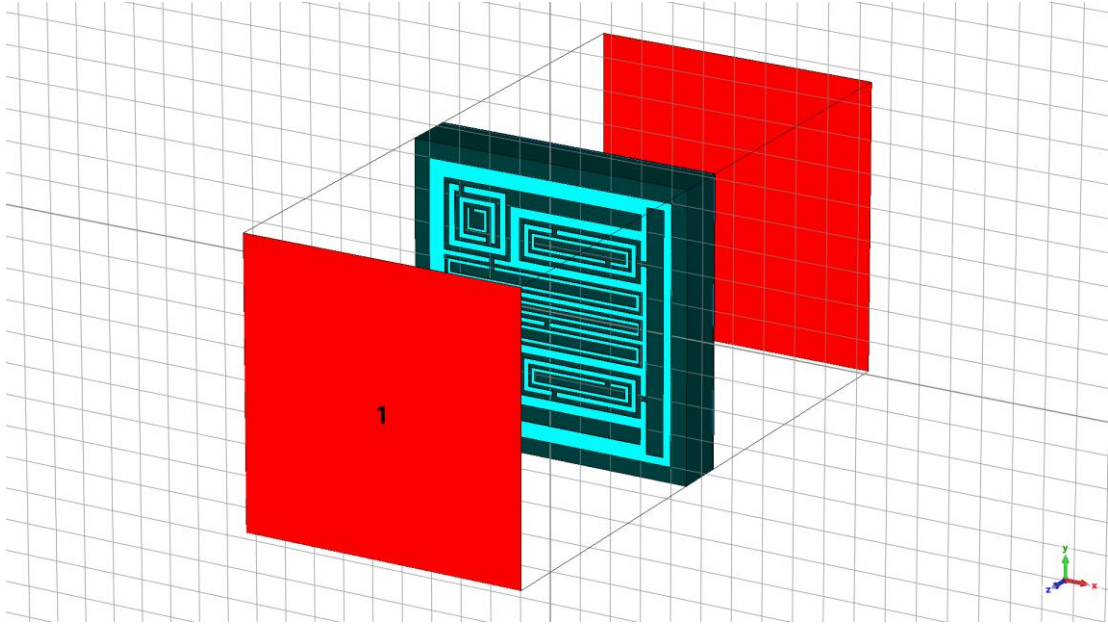


Figure 21. After applying the waveguide port.

When setting up these ports in CST, it is essential to define the correct port size and orientation. The waveguide port must cover the full cross-section of the structure to avoid errors in the simulation. If the port is too small or improperly aligned, it may result in incorrect field distributions, leading to inaccurate results. Typically, the waveguide port is defined as a rectangle or square matching the boundary of the metamaterial's input and output surfaces as shown in Figure 21. Additionally, boundary conditions play a critical role in guiding the wave. To avoid reflections at the outer boundaries, perfect electric conductor (PEC) or open boundary conditions are applied to the surrounding faces of the simulation box. The waveguide port must cover the full cross-section of the structure to avoid errors in the simulation. If the port is too small or improperly aligned, it may result in incorrect field distributions, leading to inaccurate results. Typically, the waveguide port is defined as a rectangle or square matching the boundary of the metamaterial's input and output surfaces as shown in Figure 21. Additionally, boundary conditions play a critical role in guiding the wave. To avoid reflections at the outer boundaries, perfect electric conductor (PEC) or open boundary conditions are applied to the surrounding faces of the simulation box.

This ensures that the waves are confined to travel through the metamaterial without leakage or unintended reflections. The innovative tri-layer Metal-Dielectric-Metal (MIM) structure. The innovative tri-layer Metal-Dielectric-Metal (MIM) structure. The innovative tri-layer Metal-Dielectric-Metal (MIM) structure. The innovative tri-layer Metal-Dielectric-Metal (MIM) structure.

Metal-Dielectric-Metal (MIM) structure. The innovative tri-layer Metal-Dielectric-Metal (MIM) structure. This arrangement consists of three distinct layers that work in harmony: the patterned resonator layer, which serves as the active component, a substrate film that provides support and stability, and the ground slab that acts as a foundational base. By thoughtfully varying the design of the resonator, adjusting the dimensions of the unit cell, and carefully selecting appropriate materials, one can effectively manipulate the electromagnetic characteristics of the unit.

The use of both positive and negative waveguide ports allows for the calculation of key electromagnetic parameters like the reflection coefficient (S11) and transmission coefficient (S21). The amount of energy reflected at the positive port provides insight into the absorptive properties of the metamaterial. If the metamaterial functions as a perfect absorber, the majority of the energy will be absorbed, and the reflection coefficient (S11) will be close to zero, while the transmission coefficient (S21) will also be close to zero, indicating that most of the energy has been absorbed rather than transmitted. In summary, the configuration of positive and negative waveguide ports in CST plays a fundamental role in the analysis of metamaterials. The flow of electromagnetic waves from the positive port (source) to the negative port (receiver) allows for the precise measurement of the metamaterial's absorptive, reflective, and transmissive properties. Proper alignment, port sizing, and boundary conditions are crucial for ensuring accurate and reliable simulation results.

3.6 Unit Cell Design

The design of the proposed metamaterial absorber is not only innovative but also highly practical. It offers various advantages, such as broadband absorption without the need for lumped components, insensitivity to incident angles, typical metamaterial characteristics, and a subwavelength dimension. The dimensions are 9 x 9 mm and are constructed using copper for the patch and the ground, with FR-4 as the substrate material as shown in Figure 22. To create the unit cell of the metamaterial absorber, a precisely shaped patch made of annealed copper is located at the front of the assembly, while a ground layer of annealed copper, measuring 0.035 millimeters in thickness, is positioned on the opposite side. The FR-4 substrate, which is 1.6 millimeters thick, is placed between the two copper layers. The engagement of capacitance, a key principle

in electromagnetic theory, allows for the generation of a dipole. This means that when the absorber is exposed to an electromagnetic wave, the capacitance causes a separation of charge within the absorber, creating a positive and negative pole. This promotes the distribution of surface current between the copper patch and the ground layer, enhancing the absorber's performance. In Figure 12, we present a detailed illustration of the innovative tri-layer Metal-Dielectric-Metal (MIM) structure. This arrangement consists of three distinct layers that work in harmony: the patterned resonator layer, which serves as the active component, a substrate film that provides support and stability, and the ground slab that acts as a foundational base. By thoughtfully varying the design of the resonator, adjusting the dimensions of the unit cell, and carefully selecting appropriate materials, one can effectively manipulate the electromagnetic characteristics of the unit cell. This manipulation allows researchers and engineers to achieve a tailored absorption profile, optimizing performance for specific applications. The resonator layer and ground slab are made from annealed copper, which is recognized for its ability to concentrate electric fields, flexibility, and corrosion resistance.

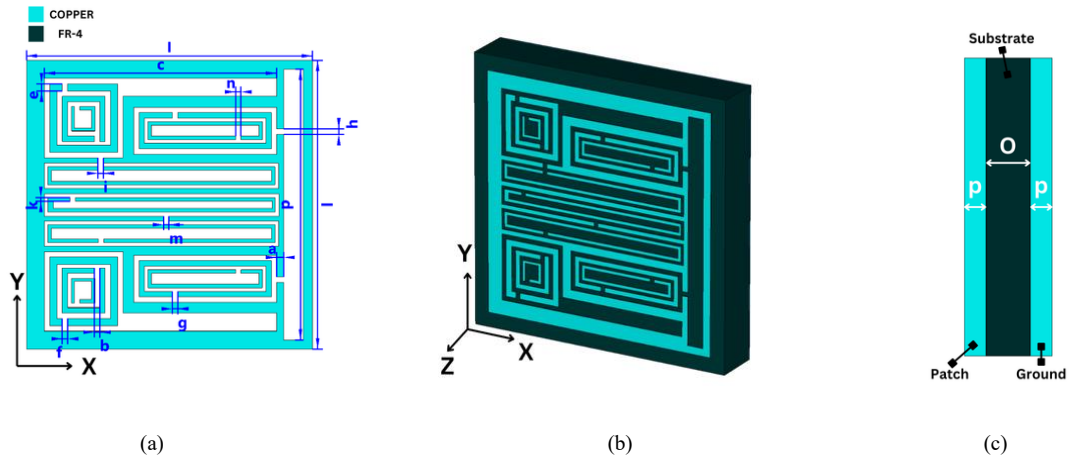


Figure 22. Design of Unit Cell - (a) Specifications of the unit cell (b) Patch and Ground View.

of the innovative tri-layer Metal-Dielectric-Metal (MIM) structure. This arrangement consists of three distinct layers that work in harmony: the patterned resonator layer, which serves as the active component, a substrate film that provides support and stability, and the ground slab that acts as a foundational base. By thoughtfully varying the design of the resonator, adjusting the dimensions of the unit cell, and carefully selecting appropriate materials, one can effectively manipulate the electromagnetic characteristics of the unit cell. This manipulation allows researchers and engineers to

achieve a tailored absorption profile, optimizing performance for specific applications. The resonator layer and ground slab are made from annealed copper, which is recognized for its ability to concentrate electric fields, flexibility, and corrosion resistance. Detailed geometrical parameters and their values are shown in Table 1. Both the ultra-thin resonator layer (p) and the ground slab (p) each have a thickness of 0.035 mm, while the substrate for the middle layer is FR-4, which has a thickness (O) of 1.6 mm. The total dimensions of the square unit cell measure $9\text{ mm} \times 9\text{ mm} \times 1.6\text{ mm}$. The patch structure illustrated in the figure is shown in Fig. 1. This design incorporates a square resonator with a side length designated as (l), positioned on an FR-4 substrate that features a copper layer for electrical connections. The resonator's architecture, which contains a series of rectangular loops and meander lines that create complex pathways, is significant. These pathways, marked with dimensions including a, b, c, d, e, f, g, and m, represent the widths and spaces of various segments in the design. The configuration of interlocking and nested paths is intended to achieve multi-band resonance characteristics by increasing the length of the pathways, thus enhancing the inductive and capacitive effects. The side view on the right reveals a cross-section of the setup, showing the copper patch atop the FR-4 substrate, separated by a dielectric layer. The labeled patch and ground plane, indicated by O and P, constitute the resonating structure. This layered configuration is not just a design choice but a functional absorber, facilitating operation across targeted frequency bands. The compact design and meticulous dimensions are crucial to achieving high absorption rates and selectivity at particular resonant frequencies.

3.7 Fundamentals of Absorption

To operate efficiently, an absorber must be able to absorb electromagnetic radiation. In order to accomplish this, it is essential to minimize the transmission and reflection of incident waves as much as possible. Consequently, the degree of absorption relies not only on the metamaterial absorber's ability to reduce transmission and reflection but also on its capability to retain energy within itself.

It's important to remember that the effectiveness of any process is directly impacted by how efficiently it is implemented, underscoring the significance of your work in the overall process as shown in Figure 23 [33]. To operate efficiently, an absorber must be able to absorb electromagnetic radiation. In order to accomplish this, it is essential.

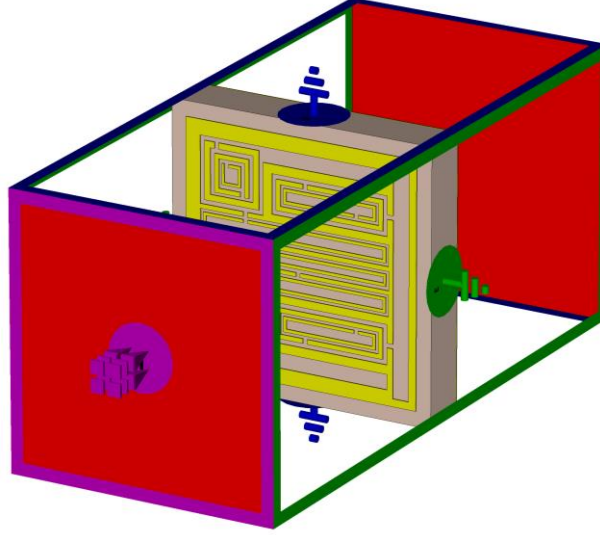


Figure 23. Port connection in the absorber.

$$A_p = 1 - r(\omega) - T(\omega) \quad (3.1)$$

The reflection symbol is represented by the letter $r(\omega)$, while $T(\omega)$ indicates the transmission coefficient, and A represents the absorption level by the absorber. However, as we know, electromagnetic waves cannot penetrate a fully conductive metal. This fact is crucial as it leads to the understanding that there will be no transmission coefficient when utilizing the whole ground. Thus, the equation becomes a key tool in our understanding of electromagnetic wave behavior,

$$A_p = 1 - r(\omega) \quad (3.2)$$

Both the impedance of the absorber and the impedance of the source of the transmitted power affect the reflection coefficient. There is no power line in this specific section of the metamaterial absorber. Instead, a waveguide port is coupled with the absorber via free space. In other words, the impedance of open space measures 377 ohms [34].

$$r(\omega) = \frac{Z_a - Z_0}{Z_a + Z_0} \quad (3.3)$$

By closely examining the equation, we can conclude that reflection will not occur when the absorber's impedance matches the open spaces. Since load and source impedance are interconnected regarding reflection impedance, ensuring a proper match is crucial and deserves significant focus [34].

$$Z_a = \sqrt{\frac{\mu_o(\omega)\mu_f(\omega)}{\epsilon_o(\omega)\epsilon_f(\omega)}} \quad (3.4)$$

The permeability of the null space and the metamaterial is denoted by the symbols μ_o and μ_f , respectively. Similarly, their permittivity is represented by ϵ_o for the null space and ϵ_f for the metamaterial. The impedance of the absorber is not a simple calculation, but a meticulous process that involves taking the square root of the ratio between its permeability and permittivity values. In contrast, the impedance of free space can be determined solely through the ratio of its permeability to permittivity. This distinction underscores the intricate nature of the properties of the metamaterial, which differ fundamentally from those of free space, and the need for precision and thoroughness in understanding their respective impedance characteristics [34].

$$Z_o = \sqrt{\frac{\mu_o(\omega)}{\epsilon_o(\omega)}} \quad (3.5)$$

3.7.1 Design Evolution

Creating broadband metamaterial absorbers for the C and X bands (with frequencies approximately ranging from 3.4 GHz to 12.4 GHz for the C band and 7.0 GHz to 11.2 GHz for the X band) is a significant endeavor for the field of electromagnetics and materials science. The process typically involves constructing a complex design of metamaterial layers capable of effectively absorbing electromagnetic waves across these frequency ranges. This is necessary since lumped elements are not involved in the design. It is essential to develop a structure comprising multiple layers, as this is the standard approach for broadband metamaterial absorbers. In order to achieve absorption across the C and X bands, each layer must demonstrate its distinct characteristics. Create a fundamental building block or unit cell that can be replicated multiple times throughout the structure.

Designing the unit cell using structures or components that resonate at various frequencies within the desired bands is advisable. Select a dielectric substrate that offers mechanical support to the structure and ensures its impedance matches that of free space within the frequency range. Incorporating substructures within the unit cell will impart resonance behavior within the C and X bands. Possible substructures include but are not limited to, cross-shaped components, split-ring resonators (SRRs), and other geometries recognized for their resonant qualities. The arrangement of these substructures within the unit cell and the spacing between them are crucial for achieving broadband absorption. Modifying the relative sizes of the substructures and

repositioning them is necessary to control the resonance frequencies accurately. It is suggested that the features and geometries of the substructures between layers be gradually altered. This gradual variation allows for the coverage of a broad frequency range without abrupt changes. Using electromagnetic modeling software may enhance the design process. Adjusting the dimensions and characteristics of the substructures facilitates more significant control over achieving properties needed for broadband absorption. When simulating the expected performance of the metamaterial structure, it is essential to focus on the C and X bands as the target frequency ranges. Building prototypes and conducting various tests will confirm that the material can absorb wideband signals. To ensure that the structure is unaffected by polarization and angle, it should be designed with additional symmetry or multiple layers if necessary for our application. When applying the absorber in practical situations, it is crucial to consider elements such as the angle of incidence and the surrounding environmental conditions. Developing broadband metamaterial absorbers without using lumped elements is complex due to the intricate design and the requirement for precise tuning. Optimization and simulation methods must be employed to attain the desired performance characteristics across the C and X bands. Experimenting with different substructure geometries and arrangement configurations is often necessary to further enhance the overall performance of the absorber.

3.7.2 The Solver List of CST Microwave Studio

CST Microwave Studio is a comprehensive software suite that facilitates microwave and radio frequency engineering and conducts electromagnetic simulations. Users are granted access to various electromagnetic analysis solvers and tools, empowering them to manipulate various aspects of the field. The Time Domain Solver (TDS) is utilized for transient simulations; the Frequency Domain Solver (FDS) is designed for frequency domain analysis; the Eigenmode Solver (EMS) employed to identify resonant frequencies and mode shapes; the Integral Equation Solver (IE), which addresses problems like scattering and radiation analysis; and the Asymptotic Solver (AS), utilized for analyzing radiation patterns and far-field results, represent some examples of the solvers in this category [35]. Specific iterations of CST Microwave Studio come with extra features, including the Thermal Solver (TS) for thermal assessments, the Particle-in-Cell Solver (PIC) for dynamics involving charged particles, and optimization tools to enhance design performance. These features are part of the

software package. The CST Microwave Studio software encompasses all of these functionalities. The capabilities of these solvers are integrated with meshing and pre-processing in the advanced software known as CST Microwave Studio. Consequently, this software is well-suited for a diverse range of applications that involve designing and analyzing electromagnetic and radio frequency (RF) systems [35].

3.7.3 CST Studio Suite 2021

Conducting simulations and examining multiphysics and electromagnetic phenomena is facilitated by a robust software tool, CST Studio Suite. This tool plays a crucial role in addressing complex questions related to electromagnetic fields, such as the behavior of antennas in different environments or the interaction of electromagnetic fields with thermal, mechanical, and fluid dynamics. These questions have applications across numerous sectors and industries. Here is a brief overview of CST Studio Suite in basic terms [34]:

CST Studio Suite performs sophisticated electromagnetic simulations across a wide frequency range, from DC (Direct Current) to high-frequency microwave and millimeter-wave devices. This wide frequency range allows it to model the behavior of electromagnetic fields in various scenarios, including antennas, RF/microwave circuits, waveguides, and other components. Through the multiphysics simulations integrated into CST Studio Suite, users can also explore how electromagnetic fields interact with other physical phenomena like thermal, mechanical, and fluid dynamics. It is utilized in various industries, including electronics, automotive, aerospace, healthcare, and telecommunications [32]. With CST Studio Suite, users can design and optimize various components like antennas, filters, and sensors. The software features a user-friendly graphical interface that allows users to quickly set up simulations, display results, and analyze data, making the process comfortable and straightforward.

It also offers a broad selection of visualization and post-processing capabilities. CST Studio Suite employs some of the most advanced numerical solvers to model electromagnetic field dynamics accurately. Supported solver types include the finite element method (FEM), the finite integration technique (FIT), and others. Users can conduct parametric studies and optimization efforts to refine designs and meet performance criteria. The program seamlessly integrates with popular computer-aided design (CAD) tools, allowing users to easily import and manipulate CAD models. In

academic settings, the CST Studio Suite is used for research in electromagnetics and related fields and for teaching electromagnetic theory. Enabling parallel processing allows users to leverage high-performance computing (HPC) capabilities, accelerating the simulation of large and complex models [35]. More experienced users can use scripting tools to automate and customize particular processes and simulations. With the aid of the versatile and widely adopted electromagnetic simulation tool CST Studio Suite, engineers and researchers from diverse domains can design, develop, and enhance electromagnetic systems and components. It plays a crucial role in advancing electromagnetic research, inspiring the development of innovative technologies [36].

3.7.4 Origin Pro

In the fast-paced world of science and engineering, researchers need tools that can keep up with their demands [33]. Origin Pro is one such tool, offering a comprehensive suite of data analysis and charting functionalities. It can handle data import and manipulation, statistical analysis, curve fitting, peak analysis, signal processing, image analysis, and present data in 2D and 3D formats. Its efficiency in data investigation and visualization makes it ideal for data exploration, hypothesis testing, and creating graphs and reports that meet publication standards. Furthermore, it supports various data formats and provides scripting options for automation and customization, enhancing the productivity of experts in academia, business, and research who deal with complex data on a daily basis [32].

3.7.5 ADS Circuit Simulation

Advanced Design System (ADS) is a sophisticated and comprehensive Electrical Design Automation (EDA) tool widely recognized for its robust circuit modeling and design capabilities. This advanced software is tailored to facilitate the design, analysis, and optimization of electrical circuits, with a particular emphasis on RF (Radio Frequency) and microwave applications. It provides designers with an efficient and reliable platform to model complex circuits, evaluate their performance, and refine designs for optimal functionality. One of the standout features of ADS is its extensive library of components [35]. These libraries include a broad range of passive and active circuit elements, ensuring that designers have access to a vast array of resources for building and testing their designs. The software also boasts an intuitive interface for schematic capture, which simplifies the design process. This user-friendly interface

allows engineers to create and modify circuit schematics effortlessly, significantly improving design efficiency and reducing the likelihood of errors during the modeling process. ADS supports advanced design exploration through its ability to perform parameter sweeps and optimization. These features allow designers to explore different design configurations, assess the impact of varying parameters, and identify the optimal setup for their circuits [35]. This capability is particularly valuable in RF and microwave circuit design, where precise parameter tuning is critical to achieving the desired performance characteristics. Another key advantage of ADS is its integrated electromagnetic (EM) simulation capabilities. By combining EM simulation with frequency-domain and time-domain simulations, the software ensures an accurate characterization of circuit behavior under various conditions. This integrated approach allows for detailed analysis of both passive and active components, ensuring that the designs meet performance expectations and adhere to industry standards. The software also supports nonlinear analysis, which is essential for analyzing circuits operating in complex, non-linear environments often encountered in RF and microwave applications. For RF and microwave circuit designers, ADS is an indispensable tool. It enables the creation of high-performance designs by providing advanced simulation capabilities and comprehensive design support. The software is widely used across various applications, from wireless communication systems and radar technologies to IoT devices and satellite communication. Its ability to handle the unique challenges of high-frequency circuit design, combined with its powerful optimization tools, makes ADS a preferred choice for engineers seeking to push the boundaries of innovation in their fields [35].

In summary, ADS stands out as a versatile and powerful tool for electrical circuit design, offering a complete suite of functions for schematic capture, parameter exploration, and performance optimization. Its integration of advanced simulation capabilities ensures accurate characterization and reliable designs.

CHAPTER 4

RESULT ANALYSIS

4.1 Reflection coefficient (s11)

The reflection coefficient of an absorber indicates the amount of electromagnetic energy that will be reflected when it interacts with the absorber. This interaction occurs when electromagnetic energy encounters the metamaterial. Metamaterial-based absorbers are artificially created items intended to effectively absorb specific frequencies or wavelengths of electromagnetic radiation, such as microwaves or infrared light. A low reflection coefficient means that most of the incoming radiation will be absorbed by the metamaterial. Conversely, a high reflection coefficient indicates that a significant portion of the incoming radiation will be reflected. Metamaterial absorbers are utilized in various fields, including energy harvesting and stealth technologies, where reducing reflection is crucial for maximizing performance.

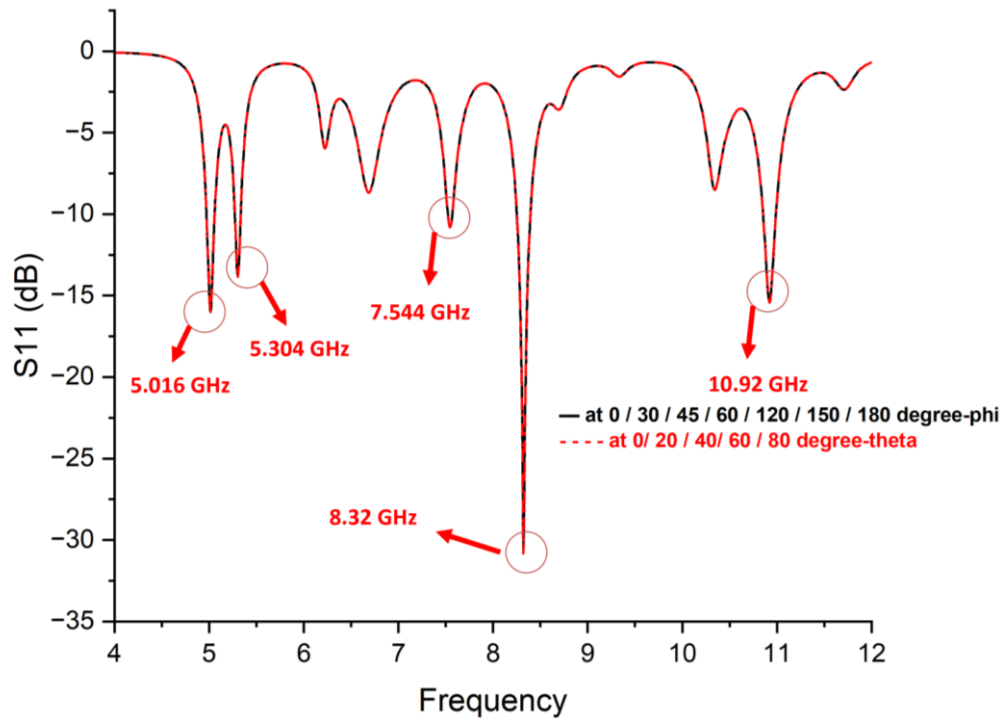


Figure 24. S11(reflection co-efficient) Parameter at different angle.

Figure 24 provides a detailed analysis of the absorption and reflectance characteristics of the proposed Microwave Metamaterial Absorber (MMA) across a wide frequency range, specifically from 4 GHz to 12 GHz. This range includes both the C-band and K-

band, highlighting the MMA's versatility. The reflection coefficient s_{11} as shown in indicates presence of five resonant frequencies, confirming its classification as "dual-band." The design of the metamaterial unit cell demonstrates precise impedance matching with that of free space as shown in Figure 24. Consequently, a significant portion of electromagnetic waves is reflected at non-resonant frequencies, resulting in reflectance values that approach perfection. At various frequencies, each split acts as a resonator when absorption reaches its maximum. Numerous modifications were implemented to the challenging ground to attain optimal broadband absorption along with negative values for permeability. It is evident from examining Fig. 14 that four resonance frequencies were successfully achieved within the C and X bands. These frequencies occurred at 5.016 GHz, 5.304 GHz, 7.544 GHz, 8.32 GHz, and 10.92 GHz, respectively. In the framework of a metamaterial model, the farfields of an absorber illustrate how electromagnetic radiation is scattered in regions that are physically separate from the absorber itself. Understanding these fields is crucial for comprehending the interaction of the absorber with incoming electromagnetic waves.

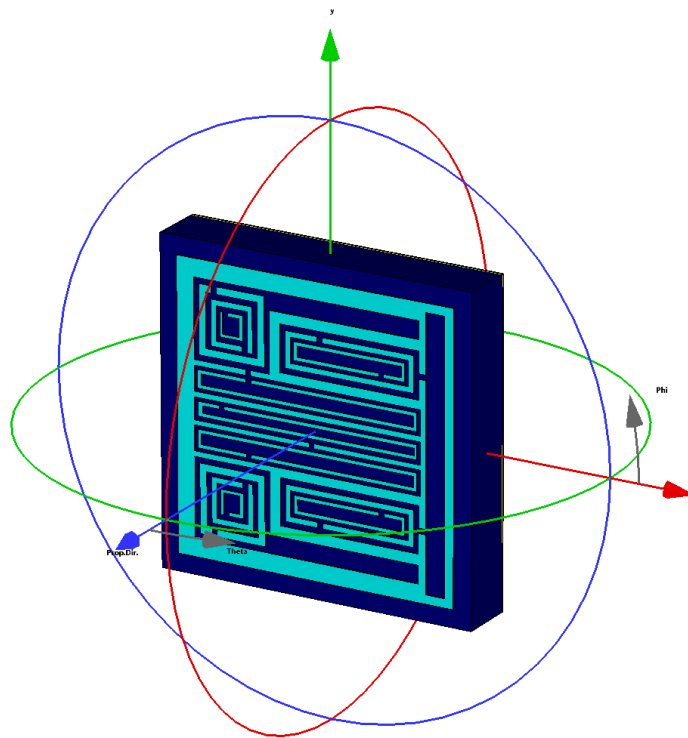


Figure 25. Fairfield.

Metamaterial absorbers can efficiently absorb and dissipate incoming radiation, converting most of the energy into heat. The absorber's effectiveness in far fields can be demonstrated by a reduction in the reflected and transmitted energy as shown in

Figure 25. This leads to a decrease in the radar cross-section and enhances signal emission or reception. This characteristic is vital for applications like stealth technology, which demands a significant decrease in radar reflections, as well as for communication systems requiring substantial signal transmission improvements. The far-field properties of metamaterial absorbers enable advancements in radar technologies, antennas, and the control of electromagnetic waves, underscoring the real-world impact of this research. These attributes are critical for evaluating metamaterial absorbers' performance and tailoring their designs to meet specific requirements.

4.2 Absorption

The remarkable phenomenon known as metamaterial absorption is made possible by a newly identified category of materials called metamaterials. These materials were purposely engineered and are commonly referred to by their specialized name. Due to their unique electromagnetic characteristics not found in natural materials, they can effectively absorb electromagnetic waves within specific frequency ranges. This capability enables their use in a wide array of applications. One of the most significant applications is in stealth technology, where metamaterials play a crucial role in reducing radar reflections [36]. By employing structures that absorb incoming radiation and dissipate the absorbed energy as heat, it is feasible to create metamaterial absorbers with nearly perfect absorption rates of one hundred percent. Creating these absorbers demands a high level of skill and craftsmanship from the builder. This technology offers a significant degree of adaptability, making it especially useful in areas such as energy harvesting, sensing, and imaging. Additionally, the technology's ability to enable precise tuning for absorption within specific frequency ranges further enhances its importance.

The metamaterial absorber exhibits remarkable absorption capabilities, particularly at its resonant frequency of 8.32 GHz, where it achieves an extraordinary absorption rate of 99.92%. This exceptional performance underscores the design's effectiveness in attenuating electromagnetic waves specifically at this frequency, making it a highly efficient material for applications requiring precise wave manipulation. In examining the graphical representation of the data, it's evident that the angle of incidence significantly influences absorption efficiency. While the absorption peaks maintain a

relatively consistent performance across a range of theta angles, discernible variations arise when considering different phi angles. This observation suggests that the absorber's effectiveness is more substantially impacted by changes in the azimuthal angle than by shifts in the polar angle, highlighting a nuanced behavior in its operational characteristics.

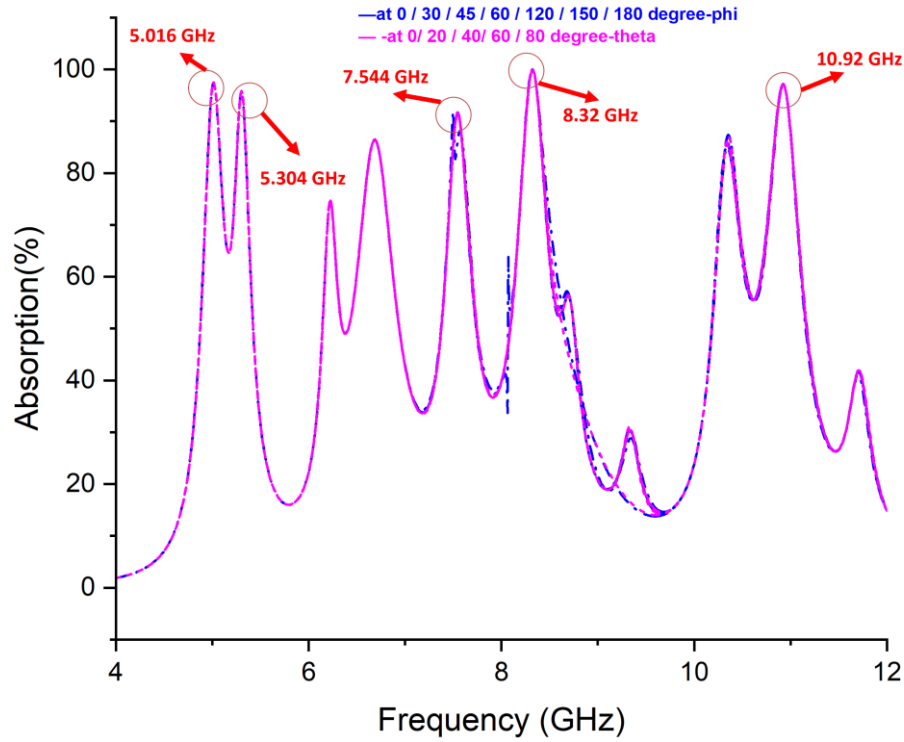


Figure 26. Absorption rate at different resonance frequency.

As shown in Figure 26 The absorption rates at various frequencies further illustrate the absorber's performance: at 8.32 GHz, it achieves an impressive 99.92%, followed by strong rates of 97.49% at 5.016 GHz, 97.14% at 10.92 GHz, 95.88% at 5.304 GHz, and 91.70% at 7.544 GHz. Together, these figures reveal a robust capacity for wave absorption across a range of frequencies, affirming the material's versatility and efficiency in practical applications.

4.3 Permeability

Examining Fig.17, we note that the permeability values at most of the resonance frequencies are almost zero, a characteristic unique to metamaterials. The permeability of a metamaterial absorber is a crucial property, contributing significantly to its remarkable absorption capabilities. Metamaterial absorbers are engineered with a unique trait- the ability to control the magnetic component of incoming electromagnetic

waves. This is achieved by designing the absorbers with reduced permeability. As a result, they can effectively manage the magnetic elements of the incoming waves, allowing for absorption over a broad frequency range.

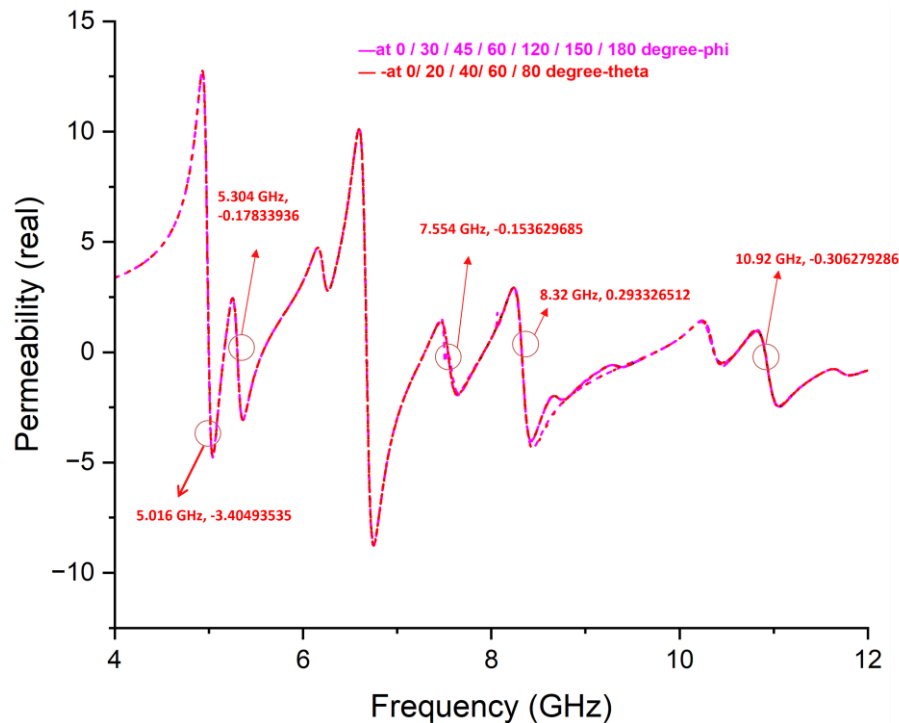


Figure 27. Change in permeability with frequency.

As shown in Figure 27 this unique trait enables the material to perform across those frequencies. Consequently, they can efficiently manage the magnetic elements of the incoming waves. Metamaterial absorbers have the potential to capture electromagnetic energy and convert it into heat effectively. This conversion is attained by modifying the material's permeability by constructing subwavelength structures and adjusting resonances. These absorbers are versatile tools for various applications, such as electromagnetic shielding, stealth technology, and efficient energy harvesting. The control over permeability and other electromagnetic characteristics facilitates this capability. The ability to produce artificial magnetic responses has expanded the range of applications for electromagnetic waves, particularly regarding their manipulation and absorption. The graphs related to reflection, absorption, and transmission in CST Microwave Studio are not just tools, but windows that provide a comprehensive overview when designing and studying metamaterial absorbers. These graphs, which are accessible within the software, offer a deep understanding of how the absorber reacts to incoming electromagnetic waves. The reflection coefficient graph, for

instance, reveals the anti-reflective characteristics of the absorber by showing the fraction of incident energy reflected from its surface. This understanding enhances the comprehension of the absorber's operation. The absorption graph, on the other hand, illustrates the absorber's ability to dissipate incoming electromagnetic energy, allowing engineers to evaluate how effectively the absorber reduces a specific frequency or a range of frequencies. This comprehensive overview provided by the graphs is a testament to the depth of understanding they offer.

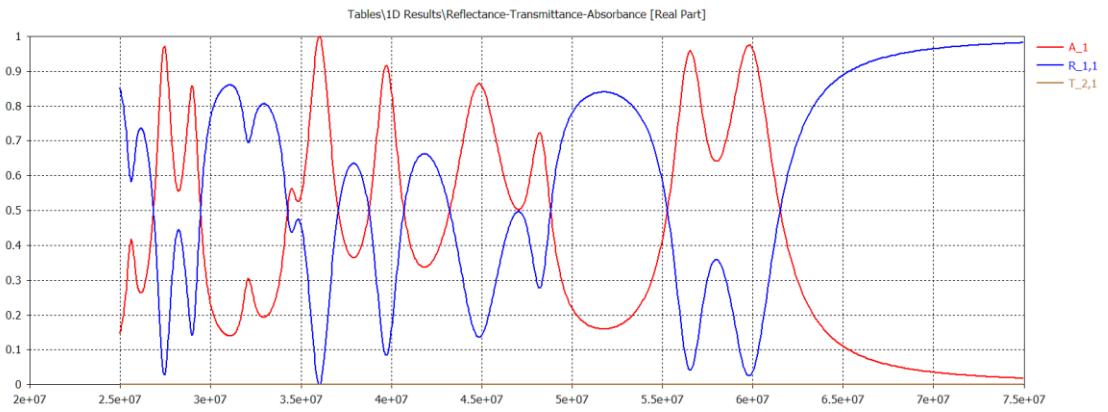


Figure 28. Reflection, absorption, and transmission.

As shown in Figure 28, the transmission coefficient graph is a valuable resource for understanding the relationship between transmission and absorption in achieving equilibrium. This graph, which represents the proportion of energy transmitted from the source to the sink, can be used to analyze this relationship. These graphs play a crucial role in the process of optimizing metamaterial absorbers for various applications. They help engineers to fine-tune the absorbers for maximum performance, including mitigating electromagnetic interference (EMI), advancing stealth technologies, and customizing absorption across multiple frequency bands. The role of these graphs in achieving performance optimization is particularly significant in the crucial application of metamaterial absorbers in mitigating EMI.

Engineers can adjust the design parameters to meet the required performance criteria for critical applications related to metamaterials, by carefully considering both sets of graphs. Figure 23, for instance, shows that the transmission reaches its lowest point where resonance is achieved, and absorption is maximized at locations similar to these points. When it comes to the process of optimizing metamaterial absorbers for a variety of applications, these graphs are a very important component. This graph, which

represents the proportion of energy transmitted from the source to the sink, can be used to analyze this relationship.

4.4 Permittivity

Understanding the characteristics and classification of metamaterial absorbers is crucial in the field of materials science. A material must contain elements that do not occur naturally to be classified as a metamaterial absorber. Typically, this means that the material shows permittivity (ϵ) or permeability (μ) values that are either negative or very close to zero. In the specific example discussed, we observe a permeability that is more significant than one that is also negative. A material is considered double negative

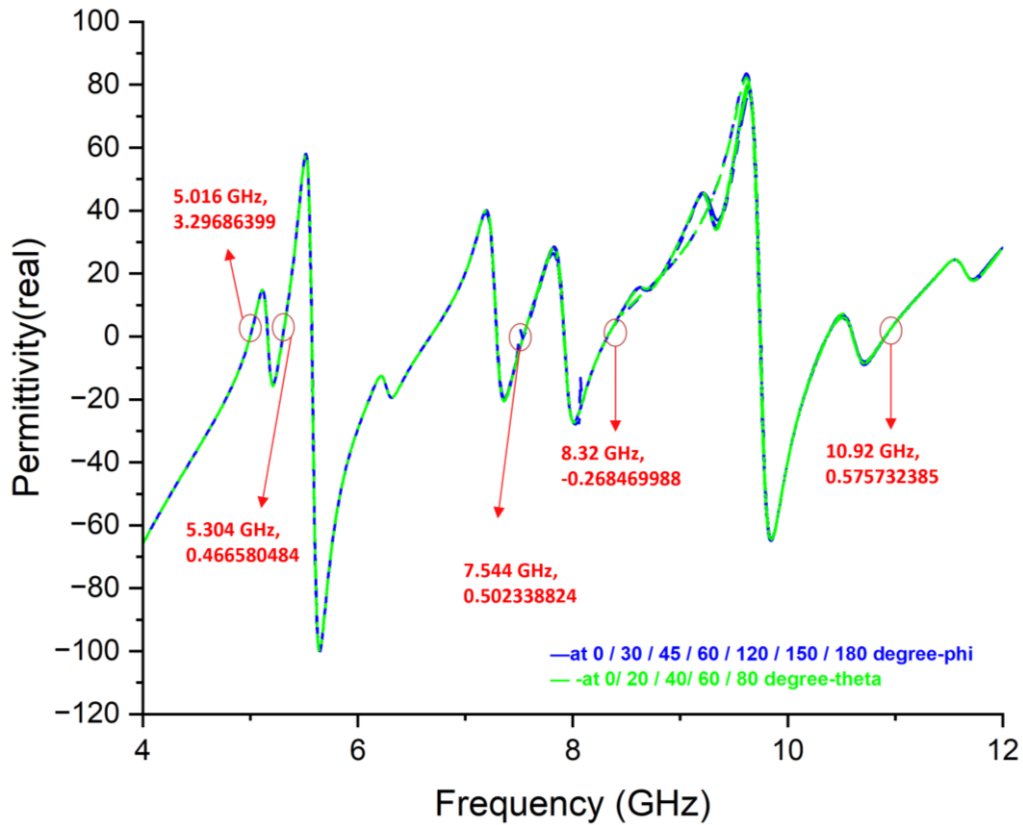


Figure 29. Change in permittivity with frequency.

if both its permittivity and permeability values are negative; such a material is referred to as "SNG" (Single Negative) material as shown in Figure 29. The proposed metamaterial (MM) absorber demonstrates a single negative value (SNG) at each resonance frequency, with permeability or permittivity values that are not just close to zero and negative but precisely so. The refractive index value reaches zero at the desired resonance frequency, a precise indication of a negative.

4.5 Refractive index

Identifying the point at which the relative permeability or permittivity values approach zero or become negative is crucial for determining the peak absorption level. This is necessary because locating this specific point is the only reliable method to gauge the amount of maximum absorption accurately. By comparing the resonance frequency data we obtained, we were able to establish that our refractive index was negative. This determination was feasible since our resonance frequency value was negative. The findings depicted in how we reached this conclusion through our research. Each resonance frequency possesses a unique negative value, often called the SNG notation, which can appear at various positions within the frequency spectrum. This notation is sometimes called the SNG notation. The MM absorber proposed for this study shows a permeability value (a) less than zero at every resonance frequency when considering the -10dB values of S11 and S21 as shown in Figure 30 .

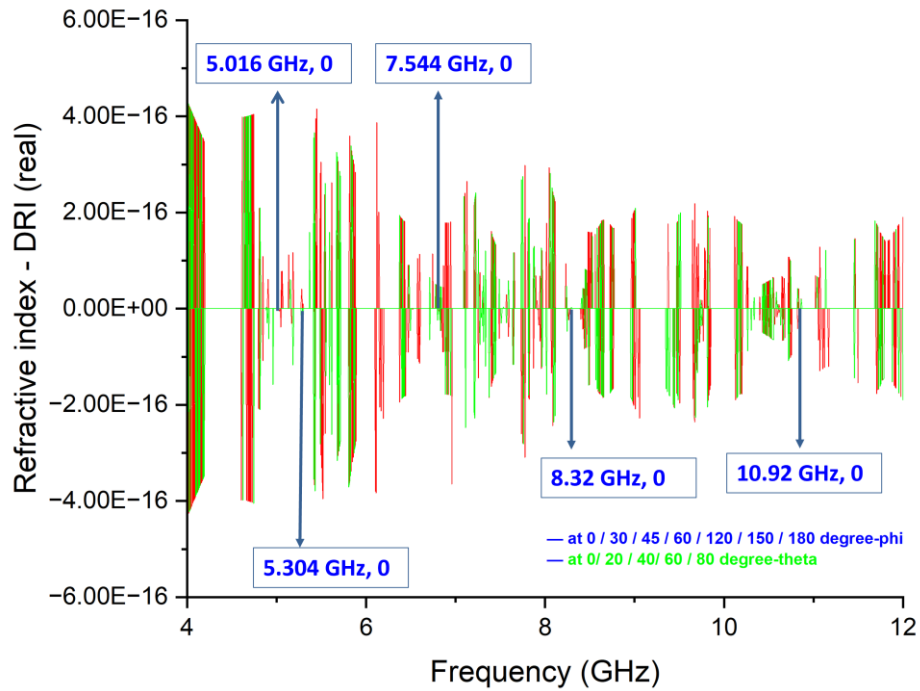


Figure 30. Refractive Index at several resonance frequencies.

This conclusion was achieved by factoring in the values of S11 and S21. Due to its permeability value (a) being lower than zero across all resonance frequencies, this absorber has been recommended for use. The refractive index of a metamaterial absorber is a crucial element that affects how electromagnetic waves interact with the absorber. The absorber's capacity to transmit and absorb electromagnetic waves

governs the interaction. Conversely, metamaterials can demonstrate unusual refractive properties, including negative or nearly zero refractive indices, unlike conventional materials that usually have positive refractive indices. This unique feature arises from the intentional design of subwavelength structures within the metamaterial. This design allows the material to bend and manipulate light or electromagnetic waves in previously unachievable ways. By precisely regulating the index of refraction during the manufacturing process, it is possible to create metamaterial absorbers capable of accurately refracting, absorbing, or altering incoming radiation. This capability is particularly beneficial in areas where conventional materials fall short, such as developing cloaking devices intended to render objects invisible to particular light or electromagnetic waves or creating superlenses for imaging beyond the diffraction limit. Both examples highlight the limitations of standard materials, underscoring the urgency and necessity of exploring metamaterials. We have gained more significant insights into optical and electromagnetic phenomena mechanisms through our ability to produce varying refractive indices in metamaterial absorbers. Furthermore, this capability has paved the way for innovative solutions across a broad spectrum of advanced technologies, facilitating their progression.

4.6 Surface current

When functioning in transverse electromagnetic (TEM) mode, the overall bandwidth of absorption is approximately 0.517, remaining consistent regardless of the angle at which the electromagnetic (EM) wave strikes. The purpose of this unique shape design is to achieve the correct ratio of capacitive and inductive components, a precision that compensates for the lack of inductive elements in the patch and significantly impacts the various resonance frequencies at 5.016 GHz, 5.304 GHz, 7.544 GHz, 8.32 GHz, and 10.92 GHz within the patch. Furthermore, the influence of damaged ground on the absorber is significant.

The damage causes the resonance frequency to shift towards the lower end of the range, highlighting the importance of maintaining the ground's integrity. The transmission characteristics through the ground also vary, and the cuts in the patch affect the reflection characteristics of the proposed metamaterial (MM) absorber. The simulation results, illustrated in Figure 31, demonstrate that the proposed absorber can achieve broadband absorption and remains insensitive to polarization at any angle. Furthermore,

as shown in Figure 31, the permittivity or permeability at the resonance frequency must be near zero (NZR) or negative for a material to exhibit metamaterial properties. This figure illustrates that the surface current is crucial in achieving the desired resonance frequencies. In addition, as demonstrated in Figure 16, in order for a material to display metamaterial qualities, its permittivity or permeability at the resonance frequency needs to be negative or close to zero (NZR). To get the right resonance frequencies, the surface current is very important, as seen in the image. In addition, as demonstrated in Figure 16, in order for a material to display metamaterial qualities, its permittivity or permeability at the resonance frequency needs to be negative or close to zero (NZR). To get the right resonance frequencies, the surface current is very important, as seen in the image.

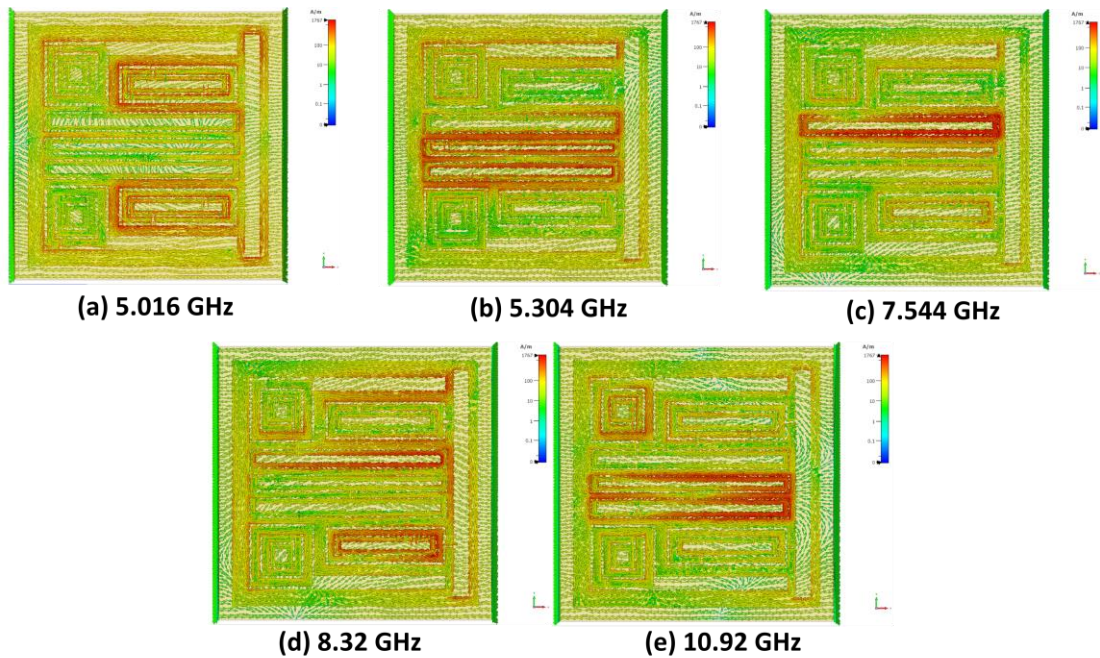


Figure 31. Surface Current at resonant frequencies.

The TABLE II provides a comparative analysis of various metamaterial absorbers, highlighting their publication year, physical dimensions, resonance frequencies, maximum absorption percentages, material characteristics, and applications. Each entry outlines a unique design and performance capability tailored to specific use cases in electromagnetic applications. The dimensions range from compact configurations like 9×9 mm (0.12λ) to larger designs such as 23×23 mm (0.41λ). Smaller dimensions are advantageous for compact devices, while larger designs may be suitable for broader applications like energy harvesting. The proposed absorber, measuring 9×9 mm

(0.12λ), is among the most compact designs, enabling it to fit into applications requiring high spatial efficiency. The resonance frequencies span a broad spectrum, demonstrating the versatility of these designs.

TABLE II. COMPARISON WITH RECENT PUBLISHED PAPERS.

Ref.	Pub. Year	Dimension	Resonance Frequencies (GHz)	Max. Absorption (%)	SNG/DNG	Applications
[22]	2024	23*23mm (0.41λ)	2.78 & 5.78	99.5	SNG	Energy Harvesting
[33]	2023	10*10 mm (0.179λ)	6.654, 8.477, and 11.3	99.84	SNG	Antenna, satellite communication, and WI-FI
[34]	2023	9*9 mm (0.12λ)	4.112, 5.768, 10.456 & 10.68	99.9	SNG	Absorber and Filter
[35]	2022	16*16mm (0.22λ)	4.196, 5.24, 8.632, 9.264, and 10.152	99.5	SNG	Sensing
[36]	2018	18*18mm (0.25λ)	4.53, 6.5, 10.2 and 11.8	99.9	Not Mentioned	RCS and EMI shielding applications
Proposed Absorber	2024	9*9 mm (0.12λ)	5.016, 5.304, 7.544, 8.32 & 10.92	99.92	SNG	N79 band 5G eMBB, Soil Sensing and Wireless and satellite communication systems

For instance, the absorber in reference [33] operates at 6.654, 8.477, and 11.3 GHz, making it effective for satellite communication and Wi-Fi. In comparison, the proposed absorber achieves resonance at 5.016, 5.304, 7.544, 8.32, and 10.92 GHz, demonstrating its capability to cover diverse applications, including N79 band 5G and soil sensing. Maximum absorption rates for all the absorbers are notably high, ranging from 99.5% to 99.92%, emphasizing their efficiency in reducing electromagnetic wave reflection and transmission. The proposed absorber, with a maximum absorption of 99.92%, surpasses the other designs in the table, showcasing its superior performance. Applications vary significantly across the absorbers, reflecting the adaptability of metamaterial technology to different needs. Reference [22] focuses on energy harvesting, leveraging broad resonance coverage for efficient energy conversion. The absorbers in references [33] and [34] cater to communication technologies and electromagnetic filtering, respectively. The proposed absorber is distinct in its

multifunctionality, supporting N79 band 5G enhanced Mobile Broadband (eMBB), soil sensing, and satellite communication systems. The single-negative (SNG) characteristic is common across most absorbers, indicating their capability to manipulate electromagnetic waves effectively. This simplicity and reliability make them suitable for a wide range of practical applications. However, the lack of a mention of SNG/DNG in reference [36] leaves the material classification unclear, highlighting a potential gap in detail. This comparison underscores the advancements in metamaterial absorber design, showcasing improvements in compactness, resonance versatility, and absorption efficiency. The proposed absorber demonstrates cutting-edge capabilities.

CHAPTER 5

CONCLUSION

5.1 Summary

A new microwave-frequency metamaterial absorber (MMA) has been designed using SRRs. This innovative configuration maximizes absorption, adjusting parameters for optimal signal absorption at various frequencies. The MMA demonstrates an impressive absorption level of 99.92% at 8.32 GHz (X-band), with a bandwidth of 143 MHz. Additionally, it achieves remarkable absorption rates of 97.49% at 5.016 GHz, 97.14% at 10.92 GHz, 95.88% at 5.304 GHz, and 91.70% at 7.544 GHz, with corresponding bandwidths of 62 MHz at 5.304 GHz, 47 MHz at 7.544 GHz, 143 MHz at 8.32 GHz, and 136 MHz at 10.92 GHz.

The performance of the metamaterial absorber is better understood by examining surface current distributions, as well as characteristics such as permittivity, permeability, and refractive index. This analysis reveals the resonant behavior of the metamaterial absorber at various frequencies. This versatile design, with its potential applications in C and X band absorption, opens up exciting possibilities for future research and development.

5.2 Conclusion

An innovative metamaterial absorber has been created, showcasing ultra-thin and ultra-wideband properties across multiple frequency bands (C and X). Notably, The MMA exhibits an impressive absorption level of 99.92% at 8.32 GHz (X-band) with a bandwidth of 143 MHz. Furthermore, absorption rates of 97.49% at 5.016 GHz, 97.14% at 10.92 GHz, 95.88% at 5.304 GHz, and 91.70% at 7.544 GHz are also noteworthy. The corresponding bandwidths for these frequencies are 62 MHz at 5.304 GHz, 47 MHz at 7.544 GHz, 143 MHz at 8.32 GHz, and 136 MHz at 10.92 GHz. To demonstrate the absorption mechanism, we utilized surface current plots and analyses of electromagnetic field distribution, providing valuable insights into the spatial distribution of the absorbed energy and the role of the metamaterial structure in achieving such high absorption rates. Additionally, our absorber displays stable absorption properties at various incident angles. The significance of angle-insensitive

metamaterial absorbers (MAs) stems from their capacity to absorb electromagnetic waves irrespective of their polarization. Their relevance is attributed to their wide-ranging applications, which include communication, radar, and imaging technologies. This versatility underscores the wide applicability of our research and its potential to promote advancements in various technological areas.

Our design considerably broadens the bandwidth compared to previously developed absorbers. This novel absorber, with its ultra-thin and ultra-wideband properties, offers a viable alternative to existing broadband absorbers and holds tremendous potential for various applications, including wireless local area networks (WLAN) and satellite communications. Its practical applications in these fields underscore the real-world impact of our research.

5.3 Achievements

The Metamaterial Absorber (MMA) has showcased remarkable capabilities by resonating at a range of frequencies, such as 5.304 GHz (62 MHz), 7.544 GHz (47 MHz), 8.32 GHz (143 MHz), and 10.92 GHz (136 MHz). Notably, its absorption rates at these frequencies surpass an exceptional threshold of 99.92%. This innovative milestone emphasizes the MMA's ability to absorb electromagnetic waves across a broad frequency spectrum. The meticulous tuning of the absorber has led to impressive performance, marking a significant leap forward in the evolution of metamaterial technology. This progress shows considerable potential for applications that demand selective and highly effective absorption across different frequency bands, demonstrating the exciting evolution in our field.

5.4 Limitation

Temperature Sensitivity: Certain metamaterial absorbers may show performance variations depending on temperature. Temperature changes can impact the properties of the materials, subsequently affecting their absorption characteristics.

Limitation on Non-Planar Surfaces: Metamaterial absorbers, typically designed for flat surfaces, are not suitable for curved or irregularly shaped surfaces.

This limitation highlights the need for alternative solutions for such surfaces. Our design considerably broadens the bandwidth compared to previously developed

absorbers. This novel absorber, with its ultra-thin and ultra-wideband properties, offers a viable alternative to existing broadband absorbers and holds tremendous potential for various applications, including wireless local area networks (WLAN) and satellite communications. Its practical applications in these fields underscore the real-world impact of our research.

Complex Design and Fabrication: The development of metamaterial absorbers with precise structures and features is a complex and costly process. Fabrication often involves advanced lithography techniques, posing significant challenges in the field.

Understanding the inherent material losses in metamaterials is crucial. This knowledge equips us to make informed decisions about their use and to develop strategies to mitigate these losses, ensuring that we are knowledgeable and well-prepared in our work with metamaterial absorbers.

5.5 Future Work

1. Integration with Complementary Technologies: The need to enhance collaboration between absorbers and other devices, such as antennas and sensors, is growing. This integration is not just a potential, but a necessity that can lead to multifunctional applications and expand absorbers' potential uses.

2. Exploration of Nano-Scale Applications: Research is underway to investigate the capabilities of miniature absorbers specifically designed for nanoelectronics. These tiny devices could drive innovations in various technological fields.

3. Adaptive Environmental Performance: The development of absorbers that can adjust to varying environmental conditions, including changes in humidity and temperature, is a crucial challenge. Overcoming this challenge could lead to significant innovations and ensure efficient operation in diverse settings.

4. Advancement of Nonlinear and Active Absorbers: There is significant potential in refining absorbers to perform complex tasks such as signal modulation and frequency conversion. This enhancement could significantly increase their versatility and functionality.

5. Bio-Medical Applications: The potential uses of absorbers in medical fields are being explored, particularly for applications in medical imaging, sensing, and

treatments. This research holds the promise of significant breakthroughs in healthcare technologies, inspiring hope for improved patient care.

6. Optimization for Space and Aerospace: Adapting absorbers for optimal performance in the challenging environments of space, which include exposure to vacuum and extreme temperature variations, is essential.

7. Enhancing Manufacturing Efficiency: Efforts are being made to improve production processes to make the fabrication of absorbers more cost-effective and scalable, allowing for broader accessibility and application.

8. Real-World Performance Testing: Conducting tests of absorbers in practical situations is not just critical, but also the final step in validating their performance and reliability. This ensures that they meet the necessary standards for efficiency in real-world applications.

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